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Bank Leverage Cycles and the External Finance Premium

By combining the approaches of Gertler and Karadi (2011) (GK) and Bernanke, Gertler, and Gilchrist (1999) (BGG), I develop a Dynamic Stochastic General Equilibrium (DSGE) model with leverage constraints both in the banking and in the nonfinancial firm sector. I calibrate this “full model” to US data. The full model matches the relative volatility of the external finance premium and the procyclicality of bank leverage and thus outperforms both a BGG and a GK-type model. For a reasonably calibrated combination of balance sheet shocks, the model reproduces a substantial share of the contraction (increase) of investment (the external finance premium) observed during the “Great Recession.”

JEL codes: E52, E53, E54

Keywords: leverage cycle, bank capital, financial accelerator, output effects of financial shocks.

THE FINANCIAL CRISIS OF 2007–2009 has drawn renewed attention to the interactions between the balance sheets of banks, the external finance premium, and economic activity. Several empirical studies find that negative shocks to the banks’ capital reduce lending and economic activity.¹ Furthermore, Adrian, Colla, and Shin (2012) document that bank leverage is procyclical. At the same time, the net worth of nonfinancial firms appears to have a positive effect on investment spending.²

I would like to thank Charles Nolan, Gregory de Walque, Hans Dewachter, Jens Iversen, Kalin Nikolov, Luca Onorante, Luc Laeven, Michael Krause, Peter Karadi, Raf Wouters, Gerard O’Reilly, Skander van den Heuvel, participants at the 2011 conference of Macroeconomic Research (MaRs) network, and two anonymous referees for valuable comments. Special thanks go to Michael Kühl, Christian Merkl, Stephane Moyon and Philipp Hartmann, and Ross Waldrop. The usual disclaimer applies.

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Received September 16, 2013; and accepted in revised form April 28, 2016.

1. See Peek and Rosengreen (1997, 2000), IMF (2010), Ciccarelli, Maddaloni, and Peydró (2011), and Fornari and Stracca (2011).

2. See Hubbard (1998) for a survey.

Journal of Money, Credit and Banking, Vol. 48, No. 8 (December 2016)

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In order to account for these features, I construct a model (which I henceforth refer to as the “full model”) where both bank and nonfinancial firm leverage matter for the cost of external funds of nonfinancial firms borrowing from banks and thus for aggregate demand. Specifically, I assume that after collecting households’ deposits, a bank may divert a fraction of its assets and declare bankruptcy, as suggested by Gertler and Karadi (2011) and will do so if the bank is not sufficiently profitable. Households’ awareness of this moral hazard problem implies that their willingness to hold deposits, and thus the bank’s ability to supply loans, depends on the present value of the bank’s expected future profits and its current net worth. Furthermore, the relationship between banks and entrepreneurs borrowing from them is subject to a “costly state verification” (CSV) problem as in the seminal financial accelerator model of Bernanke, Gertler, and Gilchrist (1999) (henceforth referred to as BGG), implying that the cost of external finance of entrepreneurs is related to their leverage.

Below, I compare the full model to a BGG-type and a Gertler–Karadi-type model. My main results can be summarized as follows: In a world with only a monetary policy and a productivity shock, the full model matches the procyclicality and volatility of bank leverage observed in the data, whereas bank leverage is strongly countercyclical and too volatile in the Gertler–Karadi-type model. The reason is that in the Gertler–Karadi-type model, the bank owns the entrepreneurs capital stock, and thus its net worth is directly exposed to the decline in the value of capital Q associated with adverse monetary policy and productivity shocks. The resulting strong on-impact decline in bank net worth implies an increase in bank leverage just as GDP declines. By contrast, in the full model, the risk of fluctuations in Q is almost completely borne by the entrepreneur, rendering *nonfinancial* firm leverage countercyclical, which is in line with the data. Bank net worth declines only marginally on impact in the full model, followed by a gradual increase triggered by higher profits on loans funded with the banks’ own net worth.

Furthermore, the BGG and, to a lesser extent, the GK model generate too low volatility of the cost of external finance relative to GDP, while the full model succeeds on this dimension. In the full model, adverse monetary policy and productivity shocks reduce future bank leverage and thus the expected profitability of banks due the gradual increase in bank net worth. The decline in expected profitability causes a cut in loan supply today, as households worried about the banks’ incentives withdraw deposits. This tightening of loan supply amplifies the increase of the external finance premium as compared to a BGG-type model, where the external finance premium is driven by entrepreneurial firm leverage alone, as well as the response of other financial variables and GDP.

Furthermore, in the full model, an adverse shock to entrepreneurial net worth causes an output contraction more than twice as large as in a BGG-type model. In the full model, the on-impact expansion of bank deposits needed to meet the increased demand for loans by entrepreneurs attempting to replace their lost funds requires an increase in bank profitability, and thus a larger increase in the external finance premium than in the BGG model. In line with the empirical evidence cited above, an adverse shock to bank net worth causes a persistent decline in GDP both in the

full model and the GK-type model. The shock decreases loan supply by individual banks as households withdraw deposits because they are worried that the decline in the value of banks increases their incentive to default and thus increases the cost of external finance. Both bank and entrepreneurial net worth shocks resemble demand shocks in that they move output and inflation in the same direction. For a reasonably calibrated combination of both net worth shocks, the model reproduces about two thirds of the trough of investment and about three quarters of the peak of the cost of external finance observed during the Great Recession associated with the financial crisis of 2007–09.

My paper differs from the emerging literature developing DSGE models with leverage constraints in both the banking and the nonfinancial firm sector in that I compare the predictions of my model regarding the cyclical properties of bank and entrepreneurial leverage as well as the cost of external finance to the data. By contrast, Hirakata, Sudo, and Ueda (2011, 2009), Gerali et al. (2010), and Dib (2010) do not attempt to match the cyclical dynamics of bank and entrepreneurial leverage. Meh and Moran (2010) match the procyclicality of bank leverage, but no predictions are made regarding the properties of the external finance premium or the leverage of nonfinancial firms. Also, by virtue of being an extension of Gertler and Karadi's (2011) model, the full model could be used to analyze unconventional monetary policy responses to financial crises considered by these authors and Gertler and Kiyotaki (2009).

Two models able to generate procyclical bank leverage have been developed by Adrian and Boyarchenko (2012) and Nuno and Thomas (2013). Their frameworks real models with leverage constraints only in the banking sector. Adrian and Boyarchenko (2012) assume an ad-hoc leverage constraint negatively affected by asset price volatility, which causes procyclical substitution between credit intermediated by banks and credit granted directly by households. In Nuno and Thomas (2013), borrowing entrepreneurs are segmented in that some are more exposed to idiosyncratic risk and have lower average returns. Due to limited liability, banks have an incentive to lend to these firms to take advantage of the upside risk. Therefore, households limit their deposit holdings and thus bank leverage. An exogenous increase in idiosyncratic risk contracts bank leverage, investment, and GDP.

Finally, my paper is also related to recent estimates of BGG-type models by Christiano et al. (2014), Fuentes-Albero (2012), and Christiano, Motto, and Rostagno (2010), who rely on shocks directly affecting the contracting problem between the entrepreneur and the bank in order to match the volatility of the cost of external finance. My results suggest that the full model's richer interactions between the real and the financial sector might be an alternative way to achieve this goal, although testing this hypothesis would require estimating the full model.

The remainder of the paper is organized as follows: Section 1 develops the model. Section 2 discusses the calibration and Section 3 compares the response of my model and a BGG-type model to the two conventional shocks. Section 4 performs the moment comparison, while Section 5 discusses the effects of shocks to the balance sheets of banks and nonfinancial firms.

1. THE MODEL

Sections 1.1–1.3 discuss the real side of the economy, while Sections 1.5 and 1.4 discuss the banking and entrepreneurial sector. The derivation of the various first-order conditions has been relegated to Appendix for reasons of brevity.

1.1 Households

The economy features a representative household with preferences described by the intertemporal utility function

$$E_t \left\{ \sum_{i=0}^{\infty} \beta^i \left[\ln(C_{t+i} - hC_{t+i-1}) - \frac{\chi}{1+\varphi} (l_{t+i}^s)^{1+\varphi} \right] \right\},$$

where C_t and l_t^s denote a Constant Elasticity of Substitution (CES) basket of consumption good varieties and labor effort, respectively, and h denotes the degree of internal habit formation. The household saves by depositing funds with banks and by buying government bonds. Both of these assets have a maturity of one quarter, yield a nominal return and, in the equilibrium considered here, are perfectly risk free in nominal terms. They are therefore perfect substitutes and earn the same interest. I denote the total financial assets of households at the end of period $t-1$ as B_{t-1}^T and the interest rate paid on these assets in period t as R_{t-1} . Households earn wage income from supplying labor to retailers and derive profit income from their ownership of retail firms and capital goods producers. Hence, their budget constraint is given by

$$P_t C_t = w_t P_t l_t + P_t \text{prof}_t + R_{t-1} B_{t-1}^T - B_t^T - P_t T_t, \quad (1)$$

where P_t , C_t , w_t , prof_t , and T_t denote the price level, consumption, the real wage, real profits, and real lump sum taxes, respectively.

1.2 Capital Goods Producers

Capital goods producers are owned by households. They produce new capital goods using a technology that yields $1 - \frac{\eta_i}{2} (\frac{I_t}{I_{t-1}} - 1)^2$ capital goods for each unit of investment expenditures I_t , with $\eta_i \geq 0$ denoting the curvature of the investment adjustment cost. Capital goods are sold to entrepreneurs at currency price $P_t Q_t$. The real expected profits of the capital goods producer are then given by

$$E_t \left\{ \sum_{i=0}^{\infty} \frac{Q_{t+i}}{Q_t} \beta^i I_{t+i} \left[Q_{t+i} \left(1 - \frac{\eta_i}{2} \left(\frac{I_{t+i}}{I_{t+i-1}} - 1 \right)^2 \right) - 1 \right] \right\},$$

where Q_t denotes the marginal utility of real income of the household.

1.3 Retailers

The varieties of goods forming the CES basket are produced by a continuum of retail firms indexed by i . Each retailer operates under monopolistic competition and is owned by households, with the demand for its product given by

$$Y_t(i) = \left(\frac{p_t(i)}{P_t} \right)^{-\varepsilon} Y_t,$$

where $\varepsilon > 1$ denotes the elasticity of substitution between varieties. Retailers hire labor $l_t(i)$ at real wage w_t from households and capital services $K_t^S(i)$ at rental rate r_t^k from entrepreneurs in economy-wide factor markets. Hence, the output of firm i is given by

$$Y_t(i) = (K_t^S(i))^\alpha (\exp(a_t)l_t(i))^{1-\alpha},$$

where a_t denotes a transitory technology shock with mean zero following an AR(1) process. I assume that retail firms have to pay fractions ψ_L and ψ_K , respectively—of their expenditures for labor and capital services in advance and borrow from banks to do so. I show in Section 1.4 that the interest rate on these loans equals the risk-free rate R_t . The loans are paid back at the end of period t . Hence, the working capital loan of retailer i $L_t^r(i)$ is given by

$$L_t^r(i) = \psi_L w_t l_t(i) + \psi_K r_t^k K_t^S(i). \quad (2)$$

This assumption helps to match the procyclicality of total loans L_t , which also include (and mainly consist of) loans to entrepreneurs, as will be discussed below. Retailers are subject to nominal rigidities in the form of Calvo (1983) contracts: Only a fraction, $1 - \xi^P$, is allowed to optimize their price in a given period. The firms that are not allowed to optimize their prices index them to past inflation at a rate γ_P and to the steady-state inflation rate Π at rate $1 - \gamma_P$. Denoting the price chosen by those firms that are allowed to optimize in period t as p_t^* , the aggregate price index evolves according to

$$P_t = \left[(1 - \xi^P) (p_t^*)^{1-\varepsilon} + \xi^P (P_{t-1} \Pi^{1-\gamma_P} (\Pi_{t-1})^{\gamma_P})^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}.$$

1.4 Banks

Following Gertler and Karadi (2011), some households in the economy are bankers. They are risk neutral and die with a fixed probability $1 - \theta$ after earning interest income on the loans they made in the previous period.³ If banker q dies, he consumes

3. I differ from Gertler and Karadi (2011) in assuming that banks are separate risk-neutral agents. Gertler and Karadi (2011) assume that banks are owned by households and transfer their terminal wealth to their household. I adopt the assumption of risk-neutral bankers because a risk-averse bank would complicate the maximization problem of the entrepreneur.

his accumulated end-of-period t real net worth $N_t^b(q)$. Dying bankers are replaced by new ones who receive a transfer N_n^b from households, which under the calibration considered is very small.⁴

Banks derive income from offering loans to nonfinancial firms. This is a key difference vis-à-vis Gertler and Karadi (2011), where banks channel funds to firms by buying equity stakes, effectively owning the firm. A banker grants two types of loans. The first type are risky interperiod loans $L_t^e(q)$ to entrepreneurs who need to buy their period $t + 1$ capital stock. These loans are due at the beginning of period $t + 1$.⁵ The second type are risk-free intraperiod working capital loans $L_t^r(q)$ to retailers who need to pay for the labor and capital services used in production, which are due at the end of period t .

Following Gertler and Karadi (2011), I assume that after collecting deposits, a banker can choose to divert some of the assets on his balance sheet and add them to his own wealth. Specifically, a banker can divert fraction $0 \leq \lambda \leq 1$ of loans to entrepreneurs and consume it. In this case, the banker declares bankruptcy and households recover the remaining assets. This implies that households will only make deposits if the banker has no incentive to default, that is, if $V_t^b(q) \geq \lambda L_t^e(q)$, where $V_t^b(q)$ denotes present value of banker q 's expected real terminal wealth:

$$V_t^b(q) = E_t \left\{ \sum_{i=0}^{\infty} (1 - \theta)^i \left(\frac{1}{\prod_{j=0}^i R_{t+1+j}^r} \right) N_{t+1+i}^b(q) \right\}, \quad R_{t+1}^r = \frac{R_t}{\Pi_{t+1}}.$$

The fact that households only make deposits if the banker has no incentive to divert assets implies that the bank will never default and household deposits will thus be risk-free in equilibrium.

By contrast, in the management of intraperiod loans, there is no moral hazard problem between bankers and depositors, and also no friction in the bank–retailer relationship. Hence, the equilibrium loan rate equals the deposit rate, implying that banks earn zero profits on these loans. The intraperiod loan business thus does not affect $N_t^b(q)$ and $V_t^b(q)$, and therefore has no impact on lending to entrepreneurs.

Let $B_t(q)$ be the amount of nominal deposits collected by the bank in order to fund interperiod loans. It follows that $P_t L_t^e(q) = P_t N_t^b(q) + B_t(q)$ and that the law

4. Christiano, Motto, and Rostagno (2010, 2014) make an analogous assumption regarding entrepreneurs in their version of the BGG model.

5. The simplifying one-period maturity assumption implies that entrepreneurial loans cannot be sold before maturity. The bank thus holds no traded assets. According to the Federal Deposit Insurance Corporation's (FDIC) Quarterly Banking Profile, the average share of assets held for trading purposes (i.e., the sum of securities available for sale and loans and leases held for sale) never exceeds 22% during the period for which such data are available (2001Q1–2013Q4). Hence, while a substantial simplification, assuming that all assets are held to maturity as in the full model appears to better approximate the behavior of FDIC-insured institutions than assuming that banks hold only traded assets as in the GK model.

of motion of banker q 's net worth is given by

$$P_t N_t^b(q) = [R_t^b P_{t-1} L_{t-1}^e(q) - R_{t-1} B_{t-1}(q)] \exp(e_t^z) \quad (3)$$

$$= P_{t-1} [(R_t^b - R_{t-1}) L_{t-1}^e(q) + R_{t-1} N_{t-1}^b(q)] \exp(e_t^z), \quad (4)$$

where R_t^b denotes the average return the bank earns on the portfolio of loans to entrepreneurs made in period $t - 1$ net of any costs associated with entrepreneurial bankruptcy. e_t^z denotes an exogenous i.i.d. shock to the capital of existing banks. The shock captures the effect of a sudden decline in the value of the assets on the bank's balance sheet for reasons unexplained by the model. Below, I will use this shock to simulate the effect of a banking crisis on the macroeconomy.

Like Gertler and Karadi (2011), I will calibrate λ such that the incentive compatibility constraint binds locally in equilibrium, hence $V_t^b(q) = \lambda L_t^e(q)$. Appendix A.3 shows that in equilibrium, all banks choose the same ratio between loans to entrepreneurs and their own net worth. Hence, $L_t^e = \phi_t^b N_t^b$, where L_t^e and N_t^b denote total loans to entrepreneurs and total bank net worth, respectively. ϕ_t^b is determined by a set of nonlinear expressions derived in Appendix A.3, which up to first order reduces to a single equation, as discussed below. In much of the discussion, I will refer to ϕ_t^b as "bank leverage" since its dynamics are both crucial for the results and are the main driver of total leverage, the ratio of total loans to bank net worth L_t/N_t^b .

N_{et}^b consists of the net worth of bankers already in business in period $t - 1$ who did not die at the beginning of period t N_{et}^b and the net worth of new bankers N_n^b , that is,

$$N_t^b = N_{et}^b + N_n^b.$$

N_{et}^b is given by

$$N_{et}^b = \theta z_{t-1,t} N_{t-1}^b, \quad (5)$$

$$z_{t-1,t} = \frac{[(R_t^b - R_{t-1}) \phi_{t-1}^b + R_{t-1}]}{\Pi_t} \exp(e_t^z), \quad (6)$$

where $z_{t-1,t}$ denotes the growth rate of the real net worth of bankers already in business in period $t - 1$ who did not die at the beginning of period t . The consumption of dying bankers is given by

$$C_t^b = (1 - \theta) z_{t-1,t} N_{t-1}^b. \quad (7)$$

For future reference, it will be useful to divide both sides of the incentive constraint $\lambda L_t^e(q) = V_t^b(q)$ by $N_t^b(q)$, which yields $\lambda \phi_t^b = \frac{V_t^b}{N_t^b}$. We will interpret $\frac{V_t^b}{N_t^b}$ as a measure of profitability, as it is the ratio of the expected value of being a banker to the own

funds of the bank as of period t that generate this value. Up to first order, this constraint can be expressed as

$$\widehat{\phi}_t^b = \left(\frac{V_t^b}{N_t^b} \right) = \sum_{i=0}^{\infty} (\theta \beta^2 z^2)^i \phi^b \frac{R^b}{R} (E_t \widehat{R}_{t+1+i}^b - \widehat{R}_{t+i}), \quad (8)$$

which may be rearranged as

$$\widehat{\phi}_t^b = E_t \left\{ \theta \beta^2 z^2 \widehat{\phi}_{t+1}^b + \phi^b \frac{R^b}{R} (\widehat{R}_{t+1}^b - \widehat{R}_t) \right\},$$

with $\widehat{\phi}_t^b = \widehat{L}_t^e - \widehat{N}_t^b$. Bank leverage thus depends positively on the expected weighted sum of profit margins on loans made in period t and after $\widehat{R}_{t+1+i}^b - \widehat{R}_{t+i}$. The intuition behind this relation is as follows: If the profit margin on loans made in period t and/or after increases, this raises the profitability of the bank ($\widehat{\frac{V_t^b}{N_t^b}}$), which, in turn, reassures depositors that the bank has no incentive to default and therefore they deposit more. Hence, the bank expands its lending to entrepreneurs and its leverage $\widehat{\phi}_t^b$. Equation (8) may therefore be interpreted as a “credit supply curve.” The difference in relation to a more conventional supply curve is that it relates the supply of loans in period t not simply to the expected profit margin on loans made in period t , but to the expected profitability of the bank and thus to the expected profit margins on both period t and future loans.

As the rearranged version of (8) illustrates, the forward looking nature of loan supply implies that given some initial value of $\widehat{\phi}_t^b$, an expected *declining path* of bank leverage requires an increase in the profit margin on bank loans (as for reasonable calibrations, $\theta \beta^2 z^2$ is very close to one). Such a declining path might be due to an on-impact increase in $\widehat{\phi}_t^b$ caused by a sudden drop in bank net worth or a sudden increase in loan demand from entrepreneurs. It might also be caused by a *gradual expected increase* in bank net worth. An expected declining path in $\widehat{\phi}_t^b$ requires an increase in the profit margin in period t loans $\widehat{R}_{t+1}^b - \widehat{R}_t$ because an equilibrium value $\widehat{\phi}_{t+1}^b < \widehat{\phi}_t^b$ implies that in period $t + 1$, profitability ($\widehat{\frac{V_{t+1}^b}{N_{t+1}^b}}$) will be lower due to a lower weighted sum of future profit margins. Those future profit margins are driven down by competition among banks trying to expand their loan supply because their own net worth is abundant relative to the loan demand they face. In particular, ($\widehat{\frac{V_{t+1}^b}{N_{t+1}^b}}$) will be lower than what is required to accommodate a leverage of $\widehat{\phi}_t^b$. Hence, given $\widehat{\phi}_t^b$, $\widehat{R}_{t+1}^b - \widehat{R}_t > 0$ is necessary to offset the shortfall in profitability. As we will become clear below, this relationship has ramifications for the response of the economy to shocks.

1.5 Entrepreneurs

Capital accumulation is carried out by risk-neutral entrepreneurs. My assumptions regarding this sector follow Christiano, Motto, and Rostagno (2010), unless otherwise stated. At the end of period t , entrepreneur j buys capital K_t^j for price $P_t Q_t$. In period $t + 1$, the entrepreneur rents part of his capital stock to retailers at a rental rate $P_{t+1} r_{t+1}^k$ and then sells the nondepreciated capital stock at price $P_{t+1} Q_{t+1}$. The average return to capital across entrepreneurs is given by

$$R_{t+1}^K = \Pi_{t+1} \frac{r_{t+1}^k + Q_{t+1}(1 - \delta)}{Q_t}. \quad (9)$$

The gross nominal return of entrepreneur j is given by $\omega_{t+1}^j R_{t+1}^K$, where ω_{t+1}^j is an idiosyncratic shock creating *ex-post* heterogeneity among entrepreneurs with a log-normal density $f(\omega^j)$, mean 1, and variance σ^2 . The entrepreneur's total assets in period $t + 1$ are thus given by $\omega_{t+1}^j R_{t+1}^K K_t^j P_t Q_t$.

To fund the acquisition of the capital stock, the entrepreneur uses his own net worth $P_t N_t^j$ and a loan $P_t L_t^j = P_t(Q_t K_t^j - N_t^j)$, which is granted by the bank at a gross nominal loan rate R_t^L . Loan and interest are paid back in period $t + 1$. Hence, a cutoff value $\bar{\omega}_{t+1}^j$ can be defined for ω_{t+1}^j such that $\bar{\omega}_{t+1}^j R_{t+1}^K P_t Q_t K_t^j = R_t^L P_t L_t^j$: for values of ω_{t+1}^j smaller than $\bar{\omega}_{t+1}^j$, the entrepreneur defaults. In case of default, the bank is entitled to seize the entrepreneur's assets $\omega_{t+1}^j R_{t+1}^K K_t^j P_t Q_t$ but has to pay a fraction μ thereof to verify their true value.

Furthermore, after the realization of $\omega_{t+1}^j R_{t+1}^K$, entrepreneurs die with a fixed probability $1 - \gamma$. Dying entrepreneurs consume their equity V_t . This assumption ensures that entrepreneurs never become fully self-financing. The fraction $1 - \gamma$ of entrepreneurs who have died are replaced by new entrepreneurs in each period who receive a transfer W^e from households, which under our calibration is very small.

At the very beginning of period $t + 1$, after the realization of aggregate uncertainty and in particular R_{t+1}^K but *before* the realization of ω^j , the expected revenue of the bank associated with a loan L_t^j is given by

$$\begin{aligned} \text{Loanrev}_{t+1}^j &= R_t^L P_t L_t^j \int_{\bar{\omega}_{t+1}^j}^{\infty} f(\omega^j) d\omega^j \\ &\quad + (1 - \mu) R_{t+1}^K P_t Q_t K_t^j \int_0^{\bar{\omega}_{t+1}^j} \omega^j f(\omega^j) d\omega^j, \end{aligned} \quad (10)$$

where the first term refers to the bank's revenue given nondefault and the second term refers to the case of default. The expected revenue associated with loan L_t^j as of period t , on the other hand, is given by

$$E_t \left\{ R_t^L P_t L_t^j \int_{\bar{\omega}_{t+1}^j}^{\infty} f(\omega^j) d\omega^j + (1 - \mu) R_{t+1}^K P_t Q_t K_t^j \int_0^{\bar{\omega}_{t+1}^j} \omega^j f(\omega^j) d\omega^j \right\},$$

where expectations are taken over R_{t+1}^K and $\bar{\omega}_{t+1}^j$.

In the previous section, I showed that given current demand for loans and the bank leverage it implies, as well as expected profit margins on loans made in future periods, the incentive compatibility constraint faced by the banker pins down the required expected return on loans made to entrepreneurs $E_t R_{t+1}^b$ (see equation (8)). Any debt contract between the entrepreneur and the bank (L_t^j, R_t^L) has to yield an expected revenue to the bank such that its expected return on these loans equals $E_t R_{t+1}^b$. Hence, the participation constraint of banks in the market for loans to entrepreneurs is given by

$$E_t \left\{ R_t^L P_t L_t^j \int_{\bar{\omega}_{t+1}^j}^{\infty} f(\omega^j) d\omega^j + (1 - \mu) R_{t+1}^K P_t Q_t K_t^j \int_0^{\bar{\omega}_{t+1}^j} \omega^j f(\omega^j) d\omega^j \right\} \quad (11)$$

$$= P_t L_t^j E_t R_{t+1}^b.$$

Hence, *in expectation*, the entrepreneurial sector bears all costs associated with bankruptcy via the loan rate.

Note that unlike in Christiano, Motto, and Rostagno's (2010) version of the BGG model, and following Zhang (2009), the loan rate is not contingent on the period $t + 1$ state but is instead determined in period t . Hence, it does not vary with the realization of R_{t+1}^K . An unexpected aggregate shock in period $t + 1$ will therefore affect the return on bank loans in period $t + 1$ via the implied change in the bankruptcy rate was not priced into the loan rate when the debt contract was made. By contrast, in Christiano, Motto, and Rostagno (2010), the loan rate varies depending on the realization of R_{t+1}^K in order to guarantee the bank a nominal return equal to the risk-free rate. In their model, the following constraint thus has to hold in every $t + 1$ aggregate state:

$$P_t L_t^j R_t = R_{t+1}^L P_t L_t^j \int_{\bar{\omega}_{t+1}^j}^{\infty} f(\omega^j) d\omega^j + (1 - \mu) R_{t+1}^K P_t Q_t K_t^j \int_0^{\bar{\omega}_{t+1}^j} \omega^j f(\omega^j) d\omega^j. \quad (12)$$

However, while adding realism to the setup, introducing nonstate contingent contracts has only minor effects on the quantitative results. Part of the reason is that by period $t + 2$, an unexpected increase in the bankruptcy rate in period $t + 1$ will be fully reflected in the loan rate due to the one-period maturity of loans.

The entrepreneur chooses the level of K_t^j and thus implicitly a combination (L_t^j, R_t^L) to maximize his expected return, which is given by

$$E_t \left\{ \int_{\bar{\omega}_{t+1}^j}^{\infty} f(\omega^j) \left(\omega^j R_{t+1}^K P_t Q_t K_t^j - R_t^L P_t L_t^j \right) d\omega^j \right\}.$$

In Appendix A.4, I show that all entrepreneurs choose the same leverage $\phi_t^e = \frac{Q_t K_t}{N_t}$, implying that $\bar{\omega}_{t+1}^j$ is the same across all firms as well, and derive the first-order

conditions. Up to first order, these equations give rise to a relationship between $E_t R_{t+1}^K / E_t R_{t+1}^b$ and the entrepreneurial leverage ratio identical to the relationship between the risk premium $E_t R_{t+1}^K / R_t$ and the leverage ratio in BGG:

$$E_t \widehat{R}_{t+1}^K - E_t \widehat{R}_{t+1}^b = \chi^l (\widehat{K}_t + \widehat{Q}_t - \widehat{N}_t), \quad (13)$$

where $\chi^l \geq 0$. Higher entrepreneurial leverage increases the probability of bankruptcy and thus expected marginal bankruptcy costs, which requires an increase in the entrepreneurial quasi-profit margin.⁶ Hence, in the full model, $E_t \widehat{R}_{t+1}^K - \widehat{R}_t$ consists of two spreads: the banking sector profit margin $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ and the entrepreneurial sector quasi-profit margin $E_t \widehat{R}_{t+1}^K - E_t \widehat{R}_{t+1}^b$. $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ is driven by bank leverage as detailed in the previous subsection, while $E_t \widehat{R}_{t+1}^K - E_t \widehat{R}_{t+1}^b$ is driven by entrepreneurial leverage.

Total entrepreneurial net worth at the end of period t consists of that part of entrepreneurial equity V_t not consumed by dying entrepreneurs and a transfer from households to entrepreneurs W^e :

$$N_t = \gamma V_t + W^e. \quad (14)$$

Entrepreneurial equity and consumption are given by

$$V_t = \left[\int_{\bar{\omega}_t}^{\infty} f(\omega^j) (\omega^j R_t^K Q_{t-1} K_{t-1} - R_{t-1}^L L_{t-1}^e) d\omega^j \right] \exp(e_t^N), \quad (15)$$

$$C_t^e = (1 - \gamma) V_t, \quad (16)$$

where e_t^N denotes an exogenous i.i.d. shock to aggregate entrepreneurial net worth.

The cutoff value of ω_t is given by

$$\bar{\omega}_t = \frac{R_{t-1}^L (Q_{t-1} K_{t-1} - N_{t-1})}{R_t^K Q_{t-1} K_{t-1}}. \quad (17)$$

Note that the lending rate is predetermined, implying that only R_t^K has a contemporaneous effect on $\bar{\omega}_t$. Finally, dividing both sides of (10) by $P_t L_t^j$, iterating one period back and using the fact that entrepreneurial leverage and the cutoff value $\bar{\omega}_t^j$ are the same across entrepreneurs as well as the law of large numbers, we have

$$\begin{aligned} R_t^b = \frac{\text{Loanrev}_t^j}{P_{t-1} L_{t-1}^j} &= \left[R_{t-1}^L \int_{\bar{\omega}_t}^{\infty} f(\omega^j) d\omega^j \right. \\ &\quad \left. + (1 - \mu) R_t^K \frac{\phi_{t-1}^e}{\phi_{t-1}^e - 1} \int_0^{\bar{\omega}_t} \omega^j f(\omega^j) d\omega^j \right], \end{aligned} \quad (18)$$

6. We call $E_t \widehat{R}_{t+1}^K - E_t \widehat{R}_{t+1}^b$ the quasi-profit margin since it does not account for the expected costs of bankruptcy, which are borne by the entrepreneur via the loan rate agreed in the debt contract.

for the average return on loans to entrepreneurs made in period $t - 1$.

1.6 Monetary Policy and Equilibrium

Monetary policy sets the risk-free interest rate, and hence the deposit rate, following an interest feedback rule of the form

$$R_t - 1 = (1 - \rho_i) \left[\begin{aligned} &R - 1 + \psi_\pi(\log(\Pi_t) - \log(\Pi)) \\ &+ \psi_y(\log(GDP_t) - \log(GDP_t^*)) \end{aligned} \right] \quad (19)$$

$$+ \rho_i(R_{t-1} - 1) + e_t^i, \quad (20)$$

where e_t^i denotes an i.i.d. monetary policy shock, where GDP_t^* denotes the natural (flexible price equilibrium) level of GDP. Following Gertler and Karadi (2011), we assume for simplicity that $\log(GDP_t) - \log(GDP_t^*)$ is proxied by the percentage deviation of real marginal cost from its steady state $\widehat{mc}_t$.

The resource constraint is given by

$$S_t = (1 - \xi^P) \left(\frac{\Pi_t}{\Pi_t^*} \right)^\varepsilon + \xi^P \left(\frac{\Pi_t}{\Pi_{t-1}^{\gamma_P} \Pi^{1-\gamma_P}} \right)^\varepsilon S_{t-1}, \quad (21)$$

$$C_t^P = C_t + C_t^e + C_t^b, \quad (22)$$

$$Y_t = S_t \left(I_t + C_t^P + \frac{R_t^K}{\Pi_t} Q_{t-1} K_{t-1} \mu \int_0^{\bar{\omega}_t} \omega f(\omega) d\omega \right), \quad (23)$$

$$Y_t = (K_{t-1})^\alpha (A_t I_t)^{1-\alpha}, \quad (24)$$

$$GDP_t = I_t + C_t + G_t, \quad (25)$$

where G_t denotes government spending. For the moment we assume that G_t is constant, that is, $G_t = G$. S_t denotes the efficiency loss arising from price dispersion. Note that S_t does not appear in the first-order approximation. The law of motion of capital is given by

$$K_t = (1 - \delta)K_{t-1} + I_t \left(1 - \frac{\eta_i}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \right), \quad (26)$$

while total loans L_t are given by

$$L_t = L_t^e + L_t^r. \quad (27)$$

1.7 Model Variants

Below, I will compare the impulse response functions and cyclical properties of the model developed above (which I refer to as the “full model”) and three alternative models. The first alternative is a BGG model with a passive banking sector. Specifically, I assume that there is no moral hazard problem between bankers and depositors ($\lambda = 0$) and that the bankers do not consume when they die.⁷ Furthermore, the loan rate on loans made in period t adjusts after period $t + 1$ shocks are realized in order to ensure that the bank receives a risk-free nominal return, which, with $\lambda = 0$, equals the risk-free rate R_t . Hence, this model features a financial accelerator as in Christiano, Motto, and Rostagno (2010) ((13) holds, but $E_t \hat{R}_{t+1}^b = \hat{R}_t$). The presence of the passive banking sector has no impact on the dynamics of the various rates of return and the real economy. However, its presence is still helpful to understand why the economy responds differently to shocks in the full model.

In the second alternative model, labeled the GK model, there is a Gertler–Karadi-type moral hazard problem between banks and depositors ($\lambda > 0$) as in the full model. However, as in Gertler and Karadi (2011), the entrepreneur funds his physical capital stock by issuing perfectly state-contingent claims to the entrepreneur’s return on capital R_t^k (defined by equation (9)), which are bought by the bank. The bank’s income therefore includes capital gains generated by changes in the value of the capital stock Q_t (just as the entrepreneur’s does in the BGG and in the full model). The entrepreneur earns zero profits state-by-state and hence accumulates no net worth. As in Gertler and Karadi (2011), there are no frictions in the bank–entrepreneur relationship. Below, I will refer to the entrepreneur’s state-contingent debt as “loans to entrepreneurs” in spite of the fact that the debt contract differs from the full model. Hence, in the GK model, we have $L_t^e = Q_t K_t$.⁸

In the third alternative model, households accumulate the capital stock K_t in order to rent it out to retailers in period $t + 1$. Hence, there are no financial frictions affecting the accumulation of physical capital.

For later reference, it is useful to repeat what constitutes the spread between the expected return on capital and the risk-free rate $E_t R_{t+1}^K - R_t$. In the full model, $E_t R_{t+1}^K - R_t = (E_t R_{t+1}^K - E_t R_{t+1}^b) + (E_t R_{t+1}^b - R_t)$; movements of the profit margin both in the entrepreneurial and in the banking sector drive $E_t R_{t+1}^K - R_t$. In the BGG model, the return on bank loans always equals the risk-free rate R_t and there is hence no profit margin in the banking sector. In the GK model, entrepreneurs earn

7. The latter assumption has a negligible effect on the results.

8. There are some small differences between my “GK” model and the model originally proposed by Gertler and Karadi (2011). Household deposits are in nominal terms in the GK model, while they are in real terms in Gertler and Karadi (2011). Furthermore, Gertler and Karadi (2011) assume variable capacity utilization. However, my results are robust against including this feature, as mentioned above.

TABLE 1
CALIBRATION OF NONPOLICY PARAMETERS

Parameter	Description	Full model	BGG model	GK model
β	Household discount factor	0.9958	0.9958	0.9958
φ	Inverse Frisch elasticity of labor supply	0.25	0.25	0.25
h	Habit formation	0.6	0.6	0.6
α	Capital elasticity of output	0.33	0.33	0.33
δ	Depreciation rate	0.025	0.025	0.025
η_i	Investment adjustment cost	4	4	4
ε	Elasticity of substitution between varieties	6	6	6
ξ^P	Probability of nonoptimization of prices	0.67	0.67	0.67
λ	Fraction of bank assets the banker can divert	0.2351	0	0.2351
θ	Survival probability of bankers	0.9915	0.9915	0.9915
N_n^b	Transfer to new bankers	0.0001	0.0012	0.0004
ψ_L	Share of retailer's labor costs paid in advance	1	1	1
ψ_K	Share of retailer's capital rental costs paid in advance	1	1	1
σ	Standard deviation of the idiosyncratic shock	0.35	0.35	—
μ	Bankruptcy costs	0.2981	0.2981	—
γ	Survival probability of entrepreneurs	0.975	0.975	—
W^e	Transfer to new entrepreneurs	0.0088	0.0107	—

zero profits as $R_t^K = R_t^b$ state by state, but $E_t R_{t+1}^b - R_t$ will be positive and time varying. Finally, in the no-friction model, $0 = \beta E_t [\frac{q_{t+1}}{q_t} \frac{R_{t+1}^K - R_t}{\Pi_{t+1}}]$, where $\beta E_t [\frac{q_{t+1}}{q_t}]$ denotes the stochastic discount factor of households, implying that up to a first-order approximation, there is thus no spread between the expected return on capital and the risk-free rate, that is, $E_t R_{t+1}^K = R_t$.

2. CALIBRATION

I calibrate the model to U.S. data over the period from 1990Q1 to 2013Q4.⁹ All data sources are described in Appendix B. After setting Π equal to the average percentage change in the GDP deflator, β is set such that the deposit rate R equals the average federal funds rate (see Table 1). Some of the parameters pertaining to the various financial frictions in the banking and entrepreneurial sector are calibrated such that the steady-state values of important financial variables in the model equal averages of certain financial data time series for the financial and nonfarm business sector. This route is also followed by Christiano, Motto, and Rostagno (2010), Meh and Moran (2010), Nolan and Thoenissen (2009), and Bernanke, Gertler, and Gilchrist (1999). Each of the targets is displayed in Table 3.

I proceed by first assuming that retailers have to fully prefinance their capital and labor costs via working capital loans, that is, $\psi_L = \psi_K = 1$. I then turn to the parameters pertaining to the entrepreneurial sector, namely, σ , μ , γ , and W^e . μ is

9. The starting year of the period is determined by the fact that the time series on risk-weighted assets (RWAs) of FDIC insured institutions I use below begins in 1990Q1.

TABLE 2
POLICY PARAMETERS AND SHOCKS

ψ_π	Coefficient on inflation in the Taylor rule	1.5
ψ_y	Coefficient on the output gap in the Taylor rule	0.5/4
ρ_i	Coefficient on the lagged interest rate in the Taylor rule	0.8
ρ_a	AR coefficient of productivity shock	0.9
σ_i	Sd. monetary policy shock	0.0016
σ_a	Sd. productivity shock	0.012

TABLE 3
IMPORTANT STEADY-STATE VALUES

Variable	Description	Value
R	Risk free rate, APR	3.97%
Π	Inflation target, APR	2.23%
$R^L - R$	Spread of the loan rate over the risk-free rate, APR	1.38%
ϕ^{e+r}	Leverage in nonfinancial firm sector	1.89
$F(\bar{\omega})$	Quarterly bankruptcy rate, percent	0.75%
$\frac{N^b}{L}$	Bank capital ratio, percent	9.55%
$\frac{G}{Y}$	Steady-state share of government in GDP	20%

set to lie in the range of estimates of bankruptcy costs cited by Carlstrom and Fuerst (1997). σ is calibrated such that the steady-state leverage of the total nonfinancial firm sector $\phi^{e+r} = \frac{N+L}{N}$ and the default rate F meet the respective target values in Table 3. The target for the probability of default is taken from Bernanke, Gertler, and Gilchrist (1999), and is close to the estimate by Christiano, Motto, and Rostagno (2014). The target for firm leverage is the ratio between total liabilities and total net worth of the nonfarm nonfinancial business sector, taken from the Flow of Funds Account (FFA) of the Federal Reserve Board.¹⁰ All data sources are described in the Appendix. γ is calibrated close to the values used by Christiano, Motto, and Rostagno (2010) and Bernanke, Gertler, and Gilchrist (1999), which allows to back the transfer to new entrepreneurs W^e .

The parameters pertaining to the banking sector are the fraction of loans the bank can divert λ , the survival probability of banks θ , and the transfer to new bankers W^b . They are calibrated to meet targets for the cost of external finance of entrepreneurs $R^L - R$, the bank capital ratio N^b/L , and the probability of bank death $1 - \theta$. The target for $R^L - R$ is an estimate of Levin, Natalucci, and Zakrajsek (2006), who estimate the cost of external finance of 796 publicly traded nonfinancial corporations over the period 1997Q1–2004Q4. They match the daily effective yield on each

10. Both net worth and total liabilities are summed up across the nonfarm business sector. The resulting nonfinancial firm leverage ratio ϕ^{e+r} is 1.89. The associated value of entrepreneurial leverage ϕ^e is 1.68, as the fraction of loans to retailers L^r in the total amounts to only 23%. Furthermore, this number is almost identical to the value of entrepreneurial leverage implied by following the procedure suggested by Fuentes-Albero (2012).

individual security issued by the firm to the estimated yield on a Treasury coupon security of the same maturity, and also correct for the differential tax treatment of government and corporate bonds. The target for N^b/L is the average ratio between tangible common equity (TCE) and RWAs of FDIC-insured institutions. Among the available empirical measures of bank net worth, TCE comes closest to the definition of bank net worth in my model. The calculation of RWA attaches weights between 0 and 1 to individual assets according to their risk and liquidity as specified by the Basel I agreement. The probability of bank death $1 - \theta$ is set close to the median probability of bank default as estimated by Carlson, King, and Lewis (2008) over the sample period.¹¹

The parameters not pertaining to the various financial frictions are calibrated according to consensus values used in the literature. The output elasticity of capital α and the depreciation rate are set to 0.33 and 0.025, respectively. The elasticity of substitution between different goods varieties ε equals 6. The probability of a retailer being unable to reoptimize its price ξ^P equals 0.67, in line with the empirical evidence of Nakamura and Steinsson (2008). The values of the parameters indexing the degree of habit formation in consumption h and the cost of adjusting investment η_i are within the range of values estimated by Smets and Wouters (2007) and Christiano, Eichenbaum, and Evans (2005). The value of the inverse Frisch elasticity of labor supply φ equals the value found by Smets and Wouters (2007) when they estimate their model variant without nominal wage stickiness.¹² For the policy rule, we use the conventional Taylor rule parameters of 1.5 for the inflation coefficient ψ_π , and 0.5 for the output gap coefficient ψ_y , along with a smoothing parameter $\rho_i = 0.8$.

In the moment comparison, I will consider two stochastic processes: a monetary policy shock and a transitory productivity shock. The standard deviation of the monetary policy shock σ_i equals 0.0016, in line with the empirical evidence by Christiano, Eichenbaum, and Evans (2005), showing that a one-standard-deviation monetary policy shock increases the policy rate R_t by 0.6 percentage points annualized (see Table 2). The persistence of total factor productivity ρ_a is set to 0.9, while the standard deviation of the productivity shock σ_a is calibrated such that the standard deviation of GDP matches the data.

The properties of the various model variants discussed in the following sections are broadly robust against a number of perturbations to the modeling setup and the calibration. These perturbations include dropping consumption habits, lowering the degree of investment adjustment cost η_i , the introduction of variable capacity utilization, nominal wage stickiness à la Erceg, Henderson, and Levin (2000), dropping advance payment for capital and labor services (i.e., setting $\psi_L = \psi_K = 0$), a lower

11. I would like to thank the authors for kindly providing an updated series covering 1990Q1–2013Q4 period.

12. The results discussed below are broadly robust against the introduction of nominal wage stickiness à la Erceg, Henderson, and Levin (2000), with a Calvo parameter for wage setting of 0.75 and an inverse Frisch elasticity of labor supply φ of 1.5.

bankruptcy cost parameter μ and setting the output response in the monetary policy rule ψ_y equal to zero. Results are available upon request.¹³

3. IMPULSE RESPONSES

I now discuss the response of the three variants of the model described in Section 1.7 to the monetary policy shock and the Total Factor Productivity (TFP) shock. All charts display percentage deviations or—in the case of rates of return and ratios—percentage point deviations of the respective variable from its steady state. The model developed in this paper is referred to as the “full model,” while the BGG model, the Gertler–Karadi model, and the model without financial frictions are labeled “BGG,” “GK,” and “nofr,” respectively. All results presented below are based on a first-order approximation of the models’ equilibrium conditions.

3.1 Monetary Policy Shock

The response of the four models to a contractionary one-standard-deviation monetary policy shock is displayed in Figures 1(a) and 1(b). The decline in GDP is much stronger in the full model than in the BGG model and the GK model, which display similar on-impact responses, while the GDP response is weakest in the no-friction model. However, the GDP decline in the GK model is more persistent than in both the BGG and in the full model. The differences in the GDP paths across the four models are mainly caused by differences in the decline in investment. The following discussion of the economic intuition for the GDP effect of the shock in the models should be read in conjunctions with the respective flow chart displayed in Figures 3(a) and 3(b), which illustrates the key mechanisms.

In all models, the increase in the central bank interest rate $\hat{R}_t$ implies that future rental income derived from capital r_t^k is discounted more heavily, which reduces the value of capital goods $\hat{Q}_t$ and thus investment. Furthermore, the increase in $\hat{R}_t$ reduces household consumption. In the three models with financial frictions, the GDP response is amplified as compared to the no-friction model due to an increase in the spread between the expected return on capital and the risk-free rate $E_t \hat{R}_{t+1}^K - E_t \hat{R}_t$, with the amplification of the the GDP response increasing in the response of the spread. I will first discuss how this increase comes about in each model, and then compare the evolution of loans and bank net worth across the three models.

Sources of the increase in $E_t \hat{R}_{t+1}^K - E_t \hat{R}_t$. In the BGG model, $E_t \hat{R}_{t+1}^K - E_t \hat{R}_t$ is driven purely by entrepreneurial leverage (see (13)) as the banking sector profit

13. Specifically, I reduce η_i to 2. In the version of the four model variants with variable capacity utilization, the elasticity of the capital utilization adjustment cost function is set equal to 1.5. In the version with nominal wage stickiness, we set the Calvo parameter for wages equal to 0.75 and the inverse Frisch elasticity of labor supply φ equal to 1.5, while the elasticity of substitution between labor varieties ε^w equals 4. The alternative value for the monitoring cost parameter μ equals 0.2.

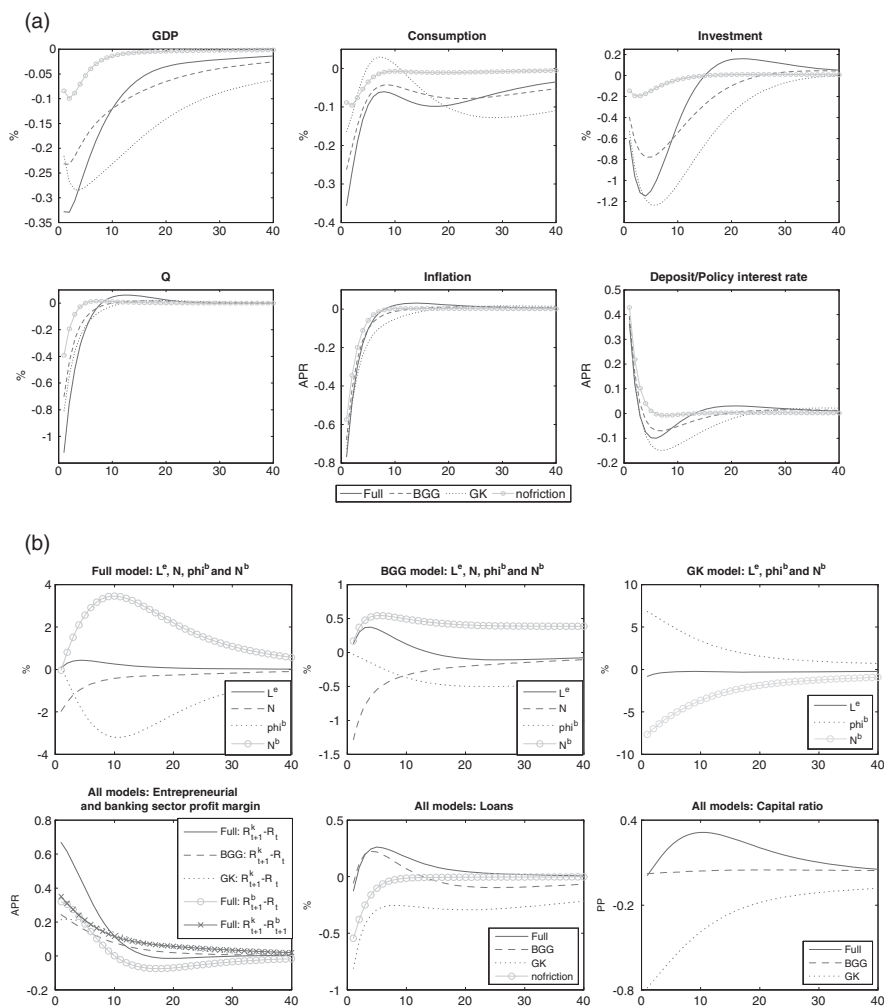


FIG. 1. (a) and (b) Monetary Tightening.

margin $R_{t+1}^b - R_t$ is always zero. The decline in $\hat{Q}_t$ lowers entrepreneurial net worth $\hat{N}_t$ (by roughly twice as much as the decline in $\hat{Q}_t$, as entrepreneurial steady-state leverage is close to 2), and thus increases entrepreneurial leverage $\phi_t^e = Q_t K_t / N_t$, as the aggregate physical capital stock responds only sluggishly to the monetary policy shock. The rise in leverage increases the probability of entrepreneurial bankruptcy; thus, expected bankruptcy costs and therefore requires an increase in $E_t \hat{R}_{t+1}^K - E_t \hat{R}_t$ via (13). Hence, $\hat{Q}_t$ and investment decline more than in the no-friction model, while entrepreneurial consumption also declines, implying that the overall drop in GDP is bigger than in the no-friction model. Furthermore, for future reference, note that in

the passive banking sector of the BGG model, bank net worth $\widehat{N}_t^b$ increases due to the increase in the monetary policy rate $\widehat{R}_t$ (Figure 1b). The hike in $\widehat{R}_t$ increases $\widehat{R}_{t+1}^b$ one for one and thus the (accounting) profits banks earn on loans they fund using their own net worth, which causes a persistent gradual increase in bank net worth and thus a persistent gradual decline in bank leverage.

In the full model, $E_t R_{t+1}^K - R_t = (E_t R_{t+1}^K - E_t R_{t+1}^b) + (E_t R_{t+1}^b - R_t)$, that is, it is the sum of two spreads. The spread between the expected return on capital and the expected return on bank loans $E_t \widehat{R}_{t+1}^K - E_t \widehat{R}_{t+1}^b$, the entrepreneurial quasi-profit margin, increases for the same reasons as in the BGG model as it is driven by entrepreneurial leverage. On top of that, and unlike in the BGG model, the declining path of bank leverage triggered by the higher risk-free rate causes an increase in the expected banking sector profit margin $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ as a consequence of the “loan supply curve” (8). A low future level of bank leverage implies that in the future, $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ is driven down by competition among banks that have abundant net worth and therefore try to expand their loan portfolio. The decline in the profit margin on future bank loans also tends to lower expected bank profitability ($\frac{V_t^b}{N_t^b}$). Hence, in period one and after, households are concerned that the bank might choose to default and withdraw deposits, thus forcing banks to restrict their supply of loans. The tightened loan supply increases $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ in period one and after (see also the discussion below equation (8)). Furthermore, rise in $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ causes steeper increase (decrease) in bank net worth (bank leverage) than in the BGG model’s passive banking sector, which enhances the on-impact jump of $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$. $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ then turns negative in quarter 10 because bank net worth (bank leverage) peaks (bottoms out), implying that bank leverage $\widehat{\phi}_t^b$ switches to an *increasing* path leading it back toward the steady state.

The on-impact increase in $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ in the full model implies that the spread between the expected return on capital and the risk-free rate $E_t \widehat{R}_{t+1}^K - \widehat{R}_t$ increases more than twice as much as in the BGG model. As a result, $\widehat{Q}_t$ and investment decline more than in the BGG model. Most of the difference in $E_t \widehat{R}_{t+1}^K - \widehat{R}_t$ between the two models is directly caused by the increase in $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ in the full model. The remainder is due to a bigger jump in the entrepreneurial profit margin $E_t \widehat{R}_{t+1}^K - E_t \widehat{R}_{t+1}^b$, which is caused by stronger drop (jump) in entrepreneurial net worth (leverage) in the full model than in the BGG model resulting from the stronger decline in $\widehat{Q}_t$.

The amplification of the monetary policy shock’s effect on GDP in the full model, which features a bank balance sheet constraint, relative to the BGG model, which does not, is in line with the euro area evidence provided by Ciccarelli, Maddaloni, and Peydró (2011). They estimate a VAR featuring survey-based measures of the change in the tightness of banks’ credit supply due to reasons related to the banks’ own balance sheets and the debtors’ balance sheets. They find that when they neutralize the effect of bank balance sheet-related changes in credit supply on GDP, thereby creating a situation comparable to the BGG model, the response of GDP to a monetary policy shock is substantially reduced.

In the GK model, the bank owns the entrepreneur, hence $R_t^K = R_t^b$. The dynamics of $E_t \widehat{R}_{t+1}^K - \widehat{R}_t$ are thus purely driven by bank leverage, and, unlike in the full model, the bank bears the risk of fluctuations in $\widehat{Q}_t$ alone. Hence, the decline $\widehat{Q}_t$ caused by the monetary tightening reduces bank net worth on impact, and, as the aggregate physical capital stock adjusts only gradually, causes an on-impact increase in bank leverage, which afterward follows a monotonously declining path. To accommodate this increase, households require an increase in bank profitability and thus an increase in $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ (equation (8)). Hence, $\widehat{Q}_t$ and investment decline more than in the no-friction model.

Note that the increase in $E_t \widehat{R}_{t+1}^K - \widehat{R}_t$ and therefore the declines in investment and output are more persistent in the GK than in the full model. This is partly because in the GK model, the declining path of bank leverage is initiated by an on-impact *jump* (caused by an on-impact decline in bank net worth caused by the on-impact drop in $\widehat{Q}_t$), implying that the return to the steady state may be achieved by a *monotonously* decreasing path. By contrast, in the full model, bank leverage starts its decreasing path directly at the steady state, implying that to return to the steady state, at some point, it has to switch to an increasing path, at which point $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ has to turn negative as well.

The evolution of loans and bank net worth. Turning first to loans, in the GK model, total loans decline due to the on-impact decline in the value of the capital stock $\widehat{Q}_t$ and the decline in economic activity. By contrast, both in the BGG and the full model, total loans increase driven by loans to entrepreneurs, a result also found by Christiano, Motto, and Rostagno (2010) in their version of the BGG model. Loans to entrepreneurs are given by $L_t^e = (Q_t K_t - N_t)$. The on-impact decline in Q_t and its subsequent recovery have largely offsetting effects on $Q_t K_t$ and N_t in absolute terms, while K_t declines only very gradually. However, on-impact N_t is further reduced by the unexpected on-impact decline in inflation, which increases the real value of $t - 1$ loans on impact (the so-called Fisher effect), which increases L_t^e . Furthermore, the subsequent recovery of N_t is slowed by the increase in the real interest rate R_{t-1}/Π_t caused by the monetary tightening, which persistently increases the cost of external funds. Therefore, $Q_t K_t$ recovers more quickly in absolute terms than N_t , which increases L_t^e further.

The prediction of the full model regarding loans differs from the models proposed by Meh and Moran (2010) and Gerali et al. (2010), who like the full-model feature leverage constraints in both the banking and the nonfinancial business sectors but predict a decline in loans to business as the GK model. However, the empirical evidence on the response of loans to a monetary tightening is mixed. While Bernanke and Blinder (1992) and Kashyap and Stein (1995) find a decrease, den Haan, Sumner, and Yamashiro (2007) find an increase based on more recent data.¹⁴

14. A recent literature has investigated whether a monetary tightening causes a change in the *quality* (as opposed to the quantity) of the loan portfolio of banks, that is, whether it lowers or increases the share of riskier loans (Landier, Thesmar, and Sraer 2011, Maddaloni and Peydo 2011, Jimenez et al. 2013).

Turning to bank net worth, following a small on-impact decline due to the unanticipated increase in the bankruptcy rate, the full model predicts an increase, as a higher risk-free rate and a higher profit margin allow net worth to grow. This prediction is shared by the model of Gerali et al. (2010) but is at odds with evidence provided by Den Haan, Sumner, and Yamashiro (2007), who find that bank net worth decreases in response to a monetary tightening. By contrast, in the GK model as well as in the model of Meh and Moran (2010), net worth decreases. Presumably, full model's prediction regarding net worth could be remedied in future research by extending the maturity of the loans to entrepreneurs, which for existing loans would increase the importance of bankruptcy-related loans and depress profit margins, and by allowing the bank to hold a fraction of its portfolio in the form of long-term traded assets, thereby exposing the bank to capital gains or losses.

3.2 Technology Shock

Figures 2(a) and 2(b) display the response of the four-model variants to a contractionary technology shock. The on-impact response of GDP is strongest in the full model, closely followed by the GK model. It is considerably weaker in the BGG and the no-friction model. As GDP declines further, the difference between the BGG and full models diminishes. The response of GDP in the no-friction model is weaker than in the full model during the first few quarters but then becomes visibly stronger. The GDP response in the GK model quickly becomes much stronger than in the full model. As before, the discussion below should be read in conjunctions with the respective flow chart displayed in Figures 3(a) and 3(b), which illustrates the key economic mechanisms driving the GDP effect of the shock.

The weaker GDP response in the BGG model than the no-friction model is due to the presence of a nominal debt contract. As also found by Christiano, Motto, and Rostagno (2010), the unexpected increase in inflation caused by the technology shock tends to reduce the debt burden of entrepreneurs. This debt reduction quickly results in a small but persistent decline of the external finance premium, thus limiting the decline in investment.

The amplification of the technology shock's effect on GDP in the full model as compared to the BGG model works in similar ways to the amplification of the monetary policy shock's effect. The anticipation of future slack in the loan market—and therefore low future profits—induces households to reduce their holdings of deposits, thus forcing individual banks to restrict their supply of loans today. The tightened loan supply increases the banking sector profit margin $E_t \hat{R}_{t+1}^b - \hat{R}_t$ (Figure 2b), thus increasing $E_t \hat{R}_{t+1}^K - \hat{R}_t$ and lowering investment and entrepreneurial consumption.

Note that both in response to a monetary policy shock and a productivity shock, bank leverage behaves procyclically in the full model but countercyclically in the GK model. In the next section, we will see that the corresponding dynamic of bank

This issue cannot be addressed in the models used here as the *ex-ante* probability of entrepreneurial default is identical across all entrepreneurs.

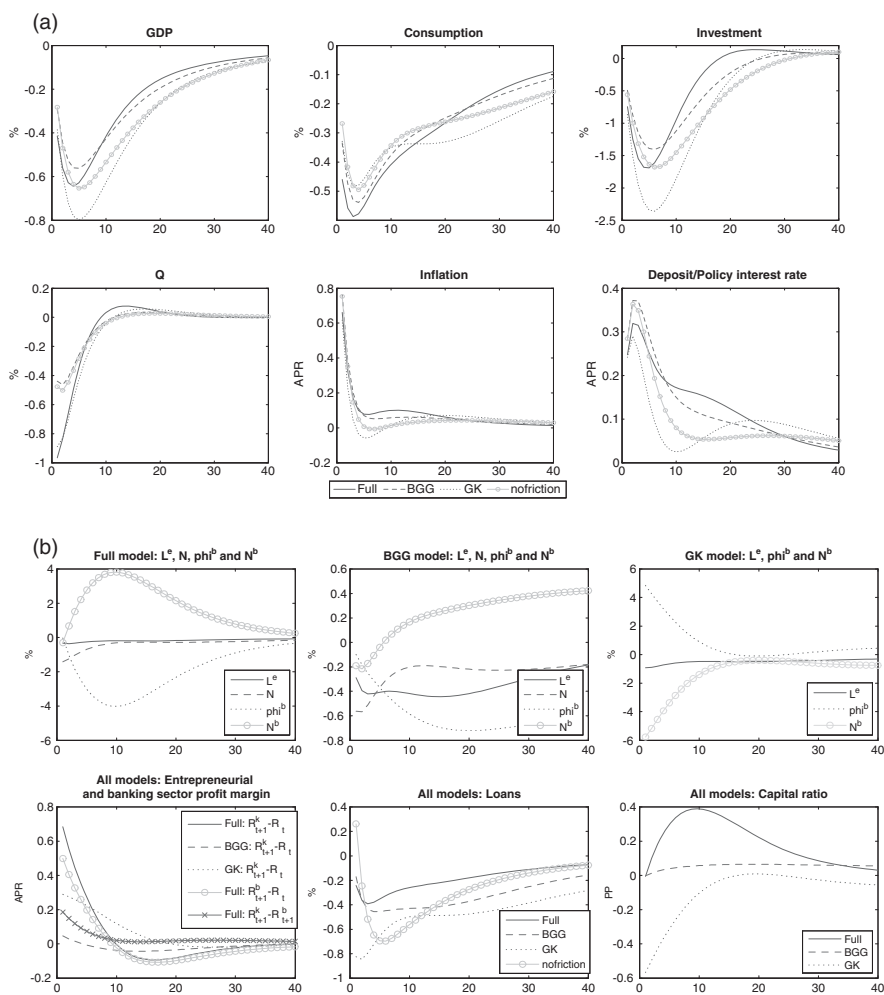


FIG. 2. (a) and (b) Adverse Productivity Shock.

leverage allows the full model to match the cyclical properties of bank leverage but prevents the GK model from doing so.

I now check whether and how the BGG model is able to generate the same response of output to monetary policy and productivity shocks if the financial accelerator χ^l is increased by raising μ , the share of a bankrupt entrepreneur's assets that has to be paid as bankruptcy cost. Setting $\mu = 1$, which is its maximum and far above available empirical estimates does indeed allow the GDP decline in the BGG model to match the one observed in the full model. However, since the BGG financial accelerator attenuates technology shocks, a higher χ^l further reduces the decline in output in

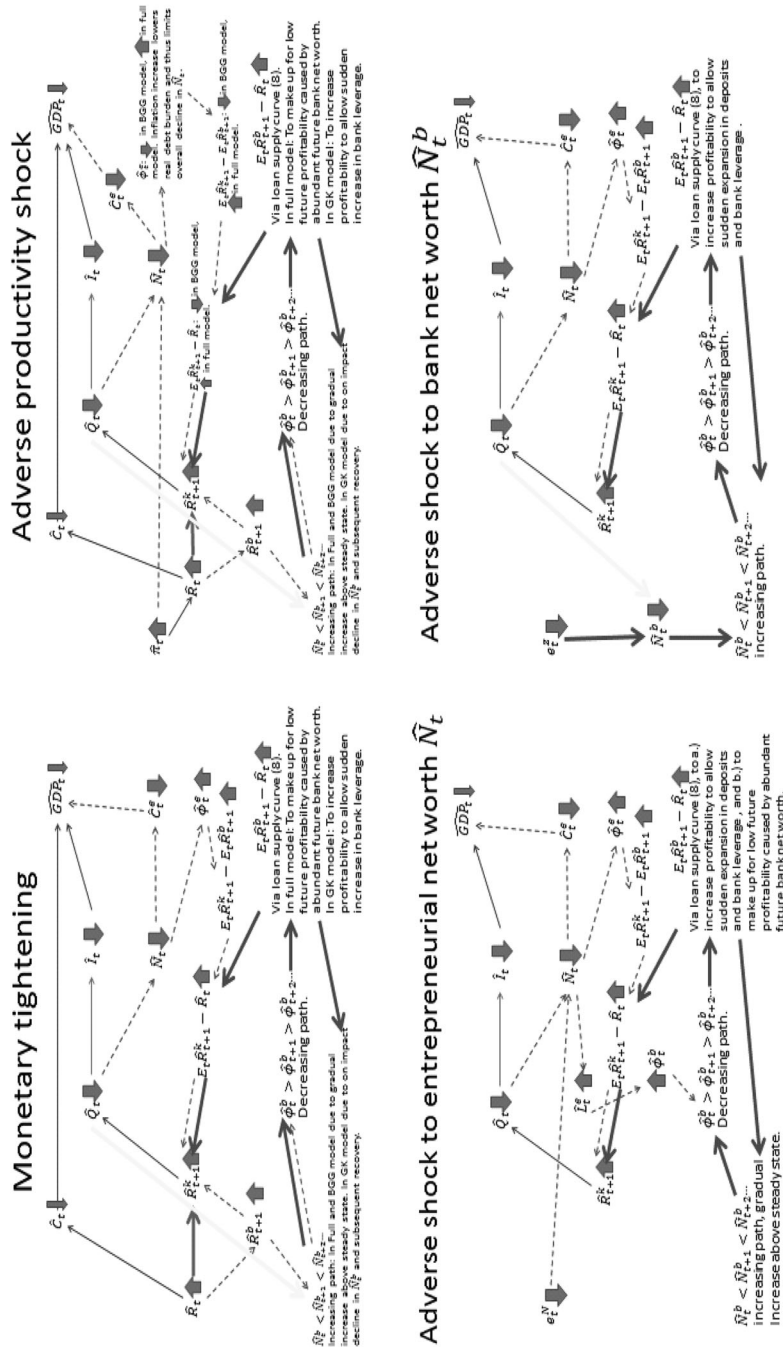
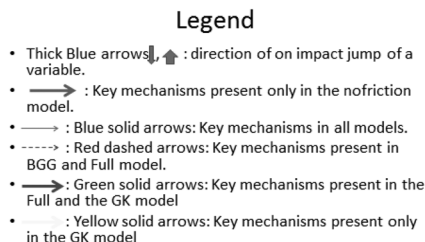


FIG. 3. (a) and (b) Flowchart Illustrating Key Economic Mechanisms.

FIG. 3. *Continued.*

response to an adverse technology shock in the BGG model relative to the full model. Moreover, $\mu = 1$ implies an unreasonably high annualized steady-state real return on capital $\frac{R^K}{\Pi} - 1$ of 17.6%.¹⁵

4. MOMENT COMPARISON

I now compare the ability of the three models with financial frictions to match the business cycle statistics of important variables in U.S. data. The real variables considered are GDP, consumption, nonresidential investment, and hours worked. The financial variables include a measure of the cost of external finance, nonfinancial firm leverage, the bank capital ratio (i.e., N_t^b/L_t), the liabilities of nonfinancial firms, bank net worth, and gross (before tax) current bank profits normalized with past net worth, which in the models is defined as z_t . In the full model, the cost of external finance is naturally given by the spread between the loan rate and the risk-free (policy) rate $R_t^L - R_t$. Although the discussion in the previous section focused on the dynamics of $E_t R_{t+1}^K - R_t$ (which is strictly speaking an unobservable variable), the impulse response functions of the two spreads are, in fact, qualitatively similar. In the BGG model, due to the state-contingent loan contract, the loan rate on loans made in period t is determined only in period $t + 1$, after the realization of aggregate shocks in period $t + 1$. I therefore choose the spread between the loan rate which borrowers expect to pay at time t in the event of nondefault and the policy rate, $E_t R_{t+1}^L - R_t$ as the BGG model's measure of the cost of external finance.¹⁶ The dynamics of this variable qualitatively resemble those of $E_t R_{t+1}^K - R_t$.¹⁷ In the GK model, the cost of external

15. An alternative way of increasing χ^I would be to increase the degree of idiosyncratic capital return uncertainty σ . However, this would also lower entrepreneurial leverage, implying that the response of the economy to a monetary policy shock is actually dampened as compared to my baseline calibration.

16. Simply using $\hat{R}_t^L - \hat{R}_t$ as the measure for the BGG model yields virtually identical results. The same is true of a measure suggested by Christiano, Motto, and Rostagno (2010): the actual transfer of resources from entrepreneurs to banks per unit of loans made minus the risk-free rate.

17. Some authors (e.g., Nolan and Thoenissen 2009) interpret $E_t \hat{R}_{t+1}^K - \hat{R}_t$ as the appropriate measure of the cost of external finance in the BGG model, although $E_t \hat{R}_{t+1}^K$ is strictly speaking not an observable

TABLE 4
STANDARD DEVIATIONS RELATIVE TO GDP

Variable	Data	Full	BGG	GK
$\widehat{GDP}_t$	1	1	1	1
$\widehat{C}_t$	0.84	0.86	0.94	0.57
$\widehat{I}_t$	4.22	3.11	2.75	3.28
$\widehat{I}_t^L$	1.66	0.92	1.19	1.0
$\widehat{R}_t^L - \widehat{R}_t, E_t \widehat{R}_{t+1}^L - \widehat{R}_t, E_t \widehat{R}_{t+1}^K - \widehat{R}_t, \text{APR}$	1.07	0.97	0.12	0.47
$\widehat{\phi}_t^{e+r}$	1.77	1.25	0.99	—
$\widehat{N}_t$	4.14	2.84	2.03	—
$\widehat{\left(\frac{N_t^b}{L_t}\right)}$	0.42	0.55	0.07	1.16
$\widehat{L}_t$	2.45	0.64	0.81	1.23
$\widehat{N}_t^b$	2.42	5.78	0.93	11.21
$\widehat{z}_t$	1.13	1.89	0.43	8.79

finance equals the spread between the expected return to capital and the risk-free rate $E_t R_{t+1}^K - R_t$.

Following Christiano, Motto, and Rostagno (2010), I measure the cost of external finance in the data as the difference between the BAA composite corporate bond rate and the effective federal funds rate.¹⁸ For the remaining variables, I use the same data employed for calculating the target values discussed in the calibration section. Both the data and the models' variables were logged (except those naturally expressed in percentage terms) and Hodrick-Prescott filtered.

Table 4 displays the standard deviations of the various variables relative to GDP. While the BGG and the GK model generate far too low volatility for the cost of external finance, the full model closely matches it. The increase in the volatility of the external finance premium as compared to the BGG model is due to the dynamics of the banking sector profit margin $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ in the full model discussed in the previous section. The higher volatility of investment, nonfinancial firm leverage $\widehat{\phi}_t^{e+r}$, and nonfinancial firm net worth $\widehat{N}_t$ in the full model has the same roots and also represents an improvement upon the BGG model.

The failure of the BGG model to match the relative volatility of the external finance premium in our moment comparison exercise is in line with recent estimates of BGG-type models by Christiano, Motto, and Rostagno (2014), Fuentes-Albero (2012), and Christiano, Motto, and Rostagno (2010), which rely on shocks directly affecting the contracting problem between the entrepreneur and the bank—for instance, shocks

variable. Using this measure does not affect the relative performance of the three models at matching the relative volatility of the cost of external finance.

18. I also considered the spread between BAA-rated bonds and the 3-month treasury bill rate, between AAA-rated bonds and the effective federal funds and between AAA-rated bonds and the 3-month treasury bill rate, used by Nolan and Thoenissen (2009). The cyclical properties of these measures of the cost of external finance differ only slightly from the difference between BAA-rated bonds and the federal funds rate.

TABLE 5
CORRELATIONS WITH GDP

Variable	Data	Full	BGG	GK
$\widehat{GDP}_t$	1	1	1	1
$\widehat{C}_t$	0.91	0.95	0.96	0.83
$\widehat{I}_t$	0.86	0.94	0.93	0.97
$\widehat{I}_t^L$	0.87	0.22	0.01	0.48
$\widehat{R}_t^L - \widehat{R}_t, E_t \widehat{R}_{t+1}^L - \widehat{R}_t, E_t \widehat{R}_{t+1}^K - \widehat{R}_t, \text{APR}$	-0.69	-0.80	-0.43	-0.84
$\widehat{\phi}_{t+r}^{e+r}$	-0.65	-0.68	-0.40	-
$\widehat{N}_t^b$	0.76	0.76	0.64	-
$\widehat{(\frac{N_t^b}{L_t})}$	-0.36	-0.40	-0.58	0.65
$\widehat{L}_t$	0.19	0.56	0.58	0.73
$\widehat{N}_t^b$	-0.2	-0.33	0.04	0.66
$\widehat{z}_t$	0.35	-0.74	-0.39	-0.05

to the degree of idiosyncratic capital return uncertainty σ , the share of bankruptcy cost μ and to entrepreneurial net worth—in order to generate the observed variation of the cost of external finance. Nolan and Thoenissen (2009) report a similar result. The full model’s richer interactions between the real and the financial sector might offer an alternative way to achieve this goal, although testing this hypothesis would require estimating the full model.

Regarding the bank balance sheet variables, the full model matches the relative volatility of the bank capital ratio $\widehat{(\frac{N_t^b}{L_t})}$ (the inverse of bank leverage), while it is much too low in the BGG and much too high in the GK model. None of the three models matches the relative volatility of bank capital $\widehat{N}_t^b$ and bank profits $\widehat{z}_t$, which are too low in the BGG model, too high in the full model and extremely high in the GK model. The high volatility of $\widehat{z}_t$, $\widehat{N}_t^b$, and $\widehat{(\frac{N_t^b}{L_t})}$ in the GK model is related to the direct impact of changes in the value of capital $\widehat{Q}_t$ on bank net worth and leverage.

Turning to the cyclicity of the various variables, note that in the data, the bank capital ratio $\widehat{(\frac{N_t^b}{L_t})}$ is countercyclical, implying procyclical bank leverage (see Table 5). The procyclicality of bank leverage is also documented by Adrian, Colla, and Shin (2012) and Adrian and Shin (2011) using alternative data sources. The full model and the BGG model are able to match this feature of the data by generating (somewhat too) procyclical bank loans, while the full model also replicates the mild countercyclicality of bank net worth, as would be expected from the net worth response to the monetary policy and productivity shock. At the same time, the countercyclicality of bank net worth is associated with countercyclical bank profits $\widehat{z}_t$, which is at odd with the data. In future research, this issue might be alleviated by extending the maturity of the loans to entrepreneurs beyond one period. With longer maturities, an unexpected increase in the bankruptcy rate would

take longer than one period to be reflected in the average interest rate on the loan portfolio because interest rates on previously made loans would remain fixed for longer, implying that bank profits would be more adversely affected. Alternatively, allowing banks to hold a fraction of its portfolio in the form of long-term traded assets would exposing its balance sheet to (likely procyclical) capital gains or losses.

By contrast, the direct and procyclical impact of changes in $\widehat{Q}_t$ on bank net worth in the GK model generates strongly procyclical bank net worth and strongly countercyclical bank leverage (a procyclical capital ratio). However, Gertler and Karadi (2013) argue that in the GK model, L_t and N_t^b are imperfect proxies of the way bank assets and net worth are measured in the data because they rely purely on market values of bank assets (i.e., the value of the physical capital stock Q_t), whereas the “Fair Value” method used to compile bank balance sheet data is a mixture of book and market values, implying both bank net worth in the data, are less procyclical than pure market values would suggest. I therefore investigate the cyclical properties of $(\frac{N_t^{b,Fair}}{L_t^{Fair}})$ and $\widehat{N}_t^{b,Fair}$, with $N_t^{b,Fair}$ and L_t^{Fair} denoting the approximation of the Fair Values of bank net worth and assets in the GK model suggested by Gertler and Karadi.¹⁹ However, in the moment comparison, these variables are procyclical, too (implying that bank leverage is still countercyclical).²⁰

An alternative way to measure the cyclicity of banking sector leverage suggested by Adrian, Colla, and Shin (2012) is to check a plot of the growth rate of bank assets against the growth rate of bank leverage for a positive correlation between the two. As shown in Figure 4, even according to this alternative measure of leverage cyclicity, the full model improves on the GK model in that bank leverage in the full model is acyclical. By contrast, in the GK model, bank leverage growth is negatively correlated with asset growth, implying that leverage is countercyclical, regardless of whether the pure market value of assets and leverage or the Fair Value method suggested by Gertler and Karadi (2013) are used. At the same time, leverage in the nonfinancial sector of the full model remains countercyclical.

Unsurprisingly, all models match the countercyclicity of the cost of external finance, although the cross-correlation is a bit too negative in the GK and the full

19. Specifically, following Gertler and Karadi (2013), in the present context, the fair value of loans L_t^{Fair} in the GK model would be given by $L_t^{Fair} = 0.75K_t + 0.25Q_tK_t + L_t'$, where $0.75K_t + 0.25Q_tK_t$ denotes the fair value of loans to entrepreneurs, while for working capital loans L_t' , book and market values are identical. Deposits D_t are calculated on market values and thus equal $D_t = L_t - N_t^b$. Finally, the fair value of equity is given by $N_t^{b,Fair} = L_t^{Fair} - D_t$. Note that in the full and BGG model, banks do not hold traded assets and thus the question of how to value them does not arise.

20. The cross-correlations with GDP amount to 0.42 and 0.46, respectively. As expected, however, $(\frac{N_t^{b,Fair}}{L_t^{Fair}})$ and $\widehat{N}_t^{b,Fair}$ are much less volatile than their market value counterparts in the GK model due to the lesser importance of fluctuations in Q_t , and thus allow the GK model to come closer to the data on this dimension. Specifically, the relative standard deviations are reduced to 0.59 and 5.31, respectively. Correspondingly, the impulse responses (not shown) are qualitatively similar to the market value counterparts but quantitatively different.

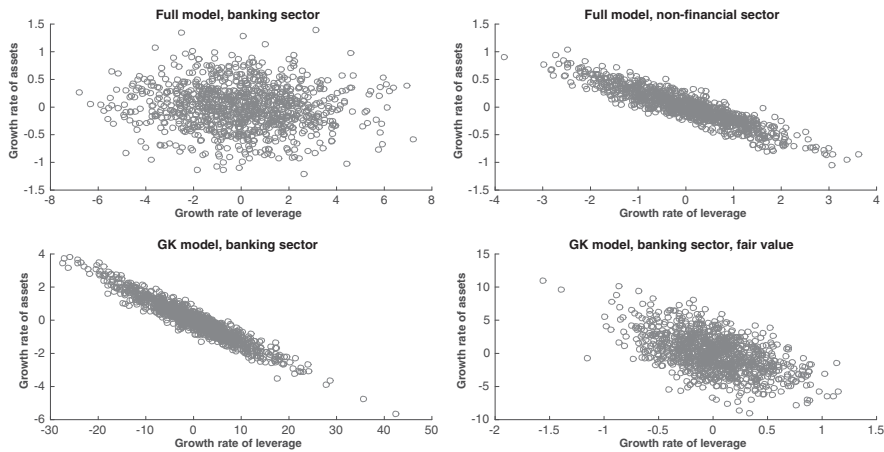


FIG. 4. Growth Rate of Assets and Leverage in the Banking and Nonfinancial Sector in the Full Model and the GK Model. The Models Were Simulated for 1,000 Periods.

TABLE 6
AUTOCORRELATIONS

Variable	Data	Full	BGG	GK
$\widehat{GDP}_t$	0.84	0.85	0.88	0.90
$\widehat{C}_t$	0.86	0.76	0.82	0.82
$\widehat{I}_t$	0.92	0.92	0.93	0.94
$\widehat{I}_t$	0.92	0.57	0.52	0.68
$\widehat{R}_t^L - \widehat{R}_t, E_t \widehat{R}_{t+1}^L - \widehat{R}_t, E_t \widehat{R}_{t+1}^K - \widehat{R}_t, APR$	0.89	0.72	0.7	0.77
$\widehat{\phi}_t^{e+r}$	0.92	0.64	0.69	—
$\widehat{N}_t$	0.92	0.61	0.60	—
$\left(\frac{N_t^b}{L_t}\right)$	0.9	0.95	0.89	0.71
$\widehat{L}_t$	0.95	0.82	0.88	0.62
$\widehat{N}_t^b$	0.7	0.95	0.89	0.7
$\widehat{z}_t$	0.68	0.62	0.52	−0.08

model, and a bit too positive in the BGG model. Loans are more procyclical than in the data in all models. The full model and the BGG model match the procyclicality of nonfinancial firm net worth and the countercyclicality of nonfinancial firm leverage, although it is perhaps a bit too low in the BGG model. Regarding the real variables, all models match the cross-correlation of $\widehat{C}_t$ and $\widehat{I}_t$ but generate a far too low cross-correlation of hours.²¹

The three models perform similarly at matching the persistence in the data (see Table 6). All models perform well at matching the autocorrelations of $\widehat{GDP}_t$, $\widehat{C}_t$, and $\widehat{I}_t$. The autocorrelation of the cost of external finance is very similar across the three

21. All models strongly improve on this dimension once nominal wage stickiness is introduced.

models but slightly too low. The same is true of the autocorrelations of $\hat{\phi}_t^{e+r}$ and $\hat{N}_t$ in the full and the BGG models, and the autocorrelation of $\hat{L}_t$ in the GK model. The autocorrelation of bank net worth is higher than in the data in the full and in the BGG model, while the GK model fails to generate any autocorrelation in bank profits.

Overall, the above discussion suggests that the amplification provided by the informational frictions in the banking–depositor relationship allows the full model to perform better than the BGG model at matching the volatility of the cost of external finance, investment, and other variables relative to output. Furthermore, the full model performs well at reproducing the statistical properties of the bank capital ratio (and thus bank leverage), which is instrumental in generating the extra volatility of the cost of external finance in the full model. By contrast, the GK model displays a procyclical bank capital ratio (countercyclical bank leverage), which is at odds with the data, as well as far too high volatility of the bank capital ratio (and thus bank leverage). While the above exercise features only two stochastic driving processes, Appendix C shows that these conclusions are largely robust to adding a number of other shocks typically considered in the DSGE literature to the models.

5. FINANCIAL SHOCKS AND CRISIS EXPERIMENT

I now examine how the model economy responds to shocks to the balance sheets of entrepreneurs and banks, and to what extent a reasonably calibrated sequence of these shocks can replicate features of the Great Recession in the U.S. economy associated with the financial crisis of 2007–09. As before, the discussion below should be read in conjunction with the respective flow chart displayed in Figures 3(a) and 3(b), which illustrates the key economic mechanisms driving the GDP effect of the shock.

Figures 5(a) and 5(b) display the response of the full model and the BGG model to a one-off -1% exogenous shock to entrepreneurial net worth $\hat{N}_t$, which I implement by setting $e_t^N = -0.01$ for one period. This type of shock has been used by numerous authors using BGG-type models, including Christiano, Motto, and Rostagno (2010) and Nolan and Thoenissen (2009). GDP declines at the trough almost twice as much in the full model as in the BGG model, due to a stronger decline in investment and entrepreneurial consumption, which is caused by a stronger increase in the spread between the expected return to capital and the risk-free rate $E_t R_{t+1}^K - R_t$.

In the BGG model, $E_t \hat{R}_{t+1}^K - \hat{R}_t$ increases because the shock directly decreases entrepreneurial net worth $\hat{N}_t$ and thus increases leverage (equation (13), recall that in the BGG model, $E_t \hat{R}_{t+1}^p = \hat{R}_t$). Hence, future rental income of entrepreneurs discounted at a higher rate, implying that the value of an additional unit of capital $\hat{Q}_t$ declines, thus enhancing the decline in $\hat{N}_t$. Hence, investment and entrepreneurial consumption decline. Furthermore, in absolute terms, the total (endogenous plus exogenous) decline in entrepreneurial net worth N_t exceeds the decline in the value

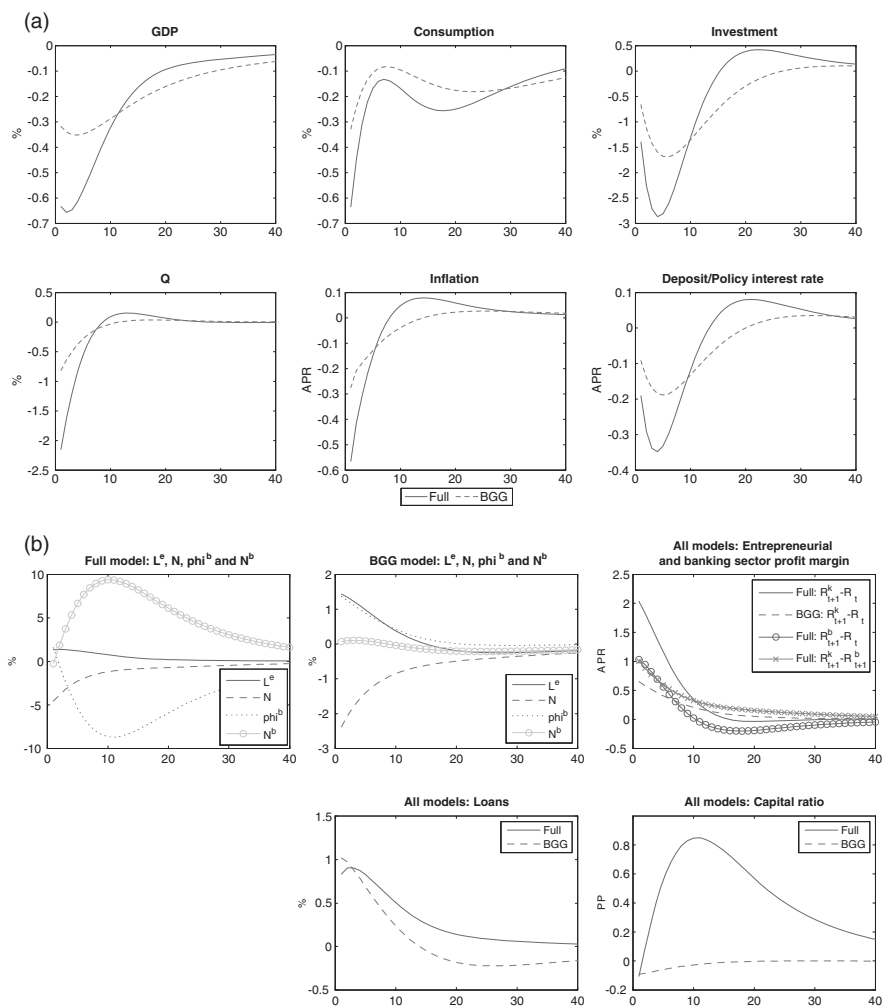


FIG. 5. (a) and (b) Adverse Entrepreneurial Net Worth Shock.

of the capital stock $Q_t K_t$, implying that entrepreneurs have to rely more strongly on loans for funding, as $L_t^e = Q_t K_t - N_t$. Therefore, deposits and leverage in the (passive) banking sector of the BGG model increase.

The same chain of events is also present in the full model. On top of that, in the full model's banking sector, loan supply is much less elastic. The on-impact expansion of bank deposits and leverage needed to meet the increased demand for loans by entrepreneurs requires an increase in the profitability ($\widehat{V_{N_t^b}^b}$) of the bank, and thus an increase in the banking sector profit margin $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ (equation(8)). Hence,

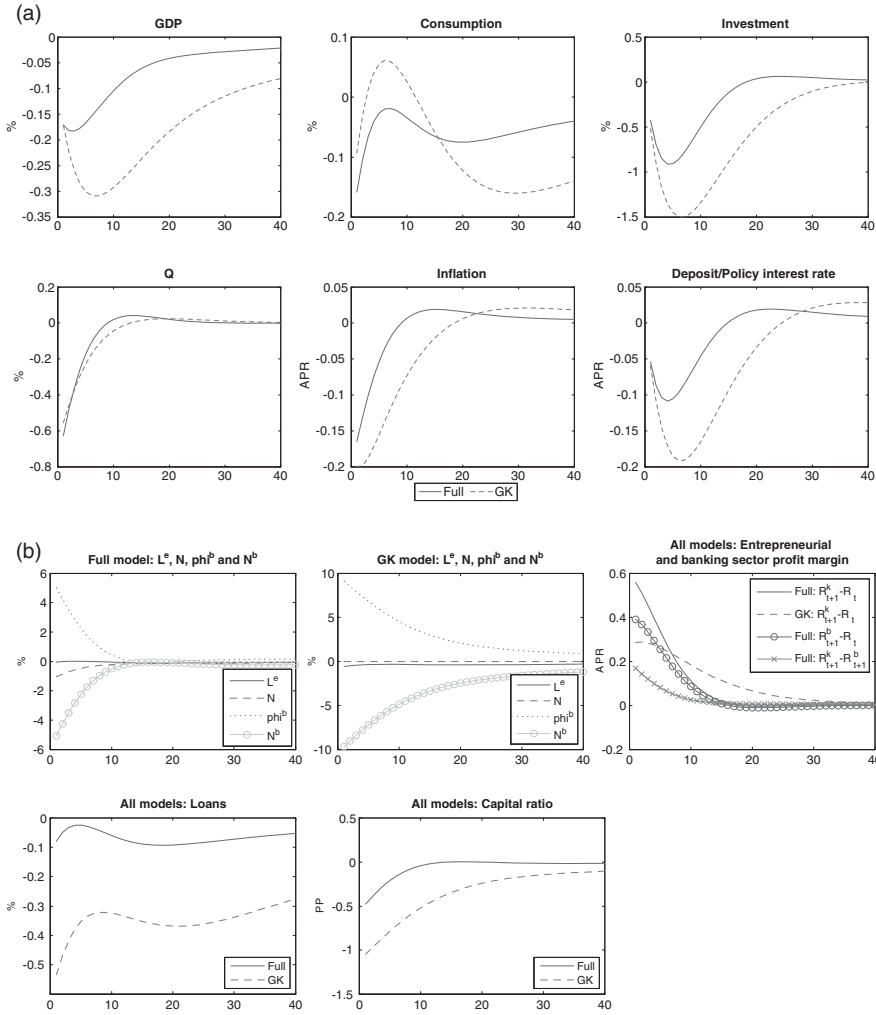


FIG. 6. (a) and (b) Adverse Bank Net Worth Shock.

$E_t R_{t+1}^K - R_t = (E_t R_{t+1}^K - E_t R_{t+1}^b) + (E_t R_{t+1}^b - R_t)$ increases by more in the full model than in the BGG model, implying a stronger drop in $\hat{Q}_t$ and thus a bigger jump in entrepreneurial leverage, which causes a further increase in $E_t \hat{R}_{t+1}^K - E_t \hat{R}_{t+1}^b$ (equation (13)). Furthermore, rise in $E_t \hat{R}_{t+1}^b - \hat{R}_t$ causes steeper increase (decrease) in bank net worth (bank leverage) than in the BGG model's passive banking sector, which enhances the on-impact jump of $E_t \hat{R}_{t+1}^b - \hat{R}_t$.

Figures 6(a) and 6(b) display the response of the full model and the GK model to a negative exogenous one-off shock of 5% to bank capital N_t^b . For that purpose, I set

$e_t^z = -0.05$ for one period. In the GK model, GDP contracts persistently chiefly due to a drop in investment. The exogenous negative shock to bank net worth increases bank leverage on impact, which requires an increase in bank profitability ($\widehat{\frac{V_t^b}{N_t^b}}$) and thus a higher profit margin $E_t \widehat{R}_{t+1}^K - \widehat{R}_t$ (recall that in the GK model, $\widehat{R}_{t+1}^b = \widehat{R}_{t+1}^K$) and thus causes an on-impact drop in $\widehat{Q}_t$, which results in a further on-impact decline in bank net worth. The increase in the banking sector profit margin $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$ implies that bank net worth gradually recovers. The decline in $\widehat{Q}_t$ lowers investment. Private consumption first decreases as the decline in bank net worth lowers the consumption of bankers (equation (7)), but this is quickly overcompensated by an increase in private household consumption in response to the decline of the central bank interest rate caused by the economic downturn.

In the full model, GDP contracts persistently due to a persistent drop in investment and entrepreneurial consumption. As in the GK model, the decline in $\widehat{N}_t^b$ increases bank leverage and thus requires an increase the profit margin on loans $E_t \widehat{R}_{t+1}^b - \widehat{R}_t$. Hence $\widehat{Q}_t$, entrepreneurial net worth (rather than bank net worth as in the GK model) and therefore investment and entrepreneurial consumption decline. The latter dominates an increase in household consumption. The capital stock and thus entrepreneurial loan demand adjust only gradually to the tighter loan supply. Therefore, total loans decline very slowly and lag significantly behind GDP and investment. The decline in investment and therefore output in the full model is lower than in the GK model partly because in the full model, the sector bearing the loss associated with the decline in $\widehat{Q}_t$ —namely, entrepreneurs—is leveraged much less than the sector bearing this risk in the GK model (namely, banks). Hence the percentage decline in entrepreneurial net worth $\widehat{N}_t$ caused by a given decline in $\widehat{Q}_t$ in the full model will be much smaller than the percentage decline in bank net worth $\widehat{N}_t^b$ caused by the same decline in $\widehat{Q}_t$ in the GK model.²²

Note that in the full model, the shock to bank capital resembles a demand shock in that it reduces both output and inflation. By contrast, in the models of Gerali et al. (2010) and Meh and Moran (2010), it appears to resemble a supply shock in that it lowers output but increases inflation. The few studies that try to empirically estimate the macroeconomic effects of a shock to bank capital record mixed results. Ciccarelli, Maddaloni, and Peydró (2011) find that, for the euro area, their proxy for a shock to bank capital moves output and inflation in the same direction. On the other hand, Fornari and Stracca (2011), using a multicountry panel VAR and a different identification scheme, also estimate that a negative shock to bank capital persistently reduces GDP, but do not find a statistically significant effect on inflation.

I now investigate whether the three models can generate an investment contraction and an increase in the cost of external finance of a magnitude similar to the experience of the U.S. economy during the Great Recession by subjecting the full model to a sequence of balance sheet shocks. In the April (2010) version of its global financial

22. Increasing the steady-state value of leverage in the entrepreneurial sector in the full model to 2.6 by assuming lower values of the degree of idiosyncratic uncertainty σ suffices to lower the trough of GDP and investment to the values reported for the GK model in Figure 4(a).

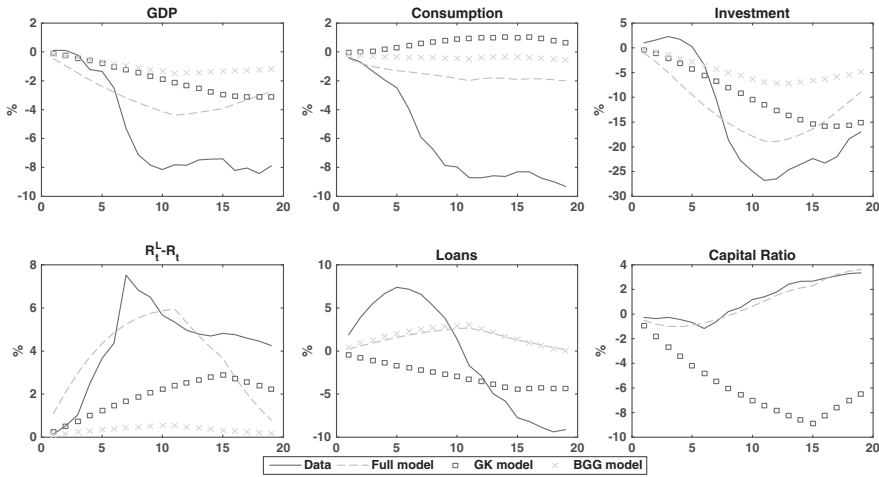


FIG. 7. Crisis Experiment.

stability report (GFS), the IMF estimates that U.S. banks had to write off 7% of the total value of the customer loans and securities on their balance sheets over the period 2007Q2–2010Q4. To assess the consequences of this type of event in the full model, I assume that banks lose an amount of their net worth equivalent to 7% of their assets. The losses are implemented by a series of 15 consecutive and equal-sized unexpected shocks to bank net worth e_t^z in the full model and in the GK model.²³ Furthermore, according to the FFAs, the net worth of nonfinancial firms in the United States also declined during the crisis. Relative to a quadratic trend, real per capita net worth declined by about 40% from 2007Q2 to 2009Q4. Therefore, I add a series of 11 consecutive unexpected shocks to entrepreneurial net worth e_t^N to the full model and the BGG model, such that given the sequence of shocks to bank net worth, the decline of entrepreneurial net worth amounts to 20% by 2009Q4. For this simulation, I use a versions of the models with nominal wage stickiness à la Erceg, Henderson, and Levin (2000) and variable capacity utilization.

Figure 7 displays the log-deviation of per capita GDP, consumption, fixed non-residential investment, and our measure of credit to nonfinancial businesses in the U.S. economy from a quadratic trend from 2007Q2 to 2011Q4, normalized by their log-deviation from trend in 2007Q1, and the deviation of our measure of the cost of external finance and the bank capital ratio from their values in 2007Q1, all labeled as “Data.”²⁴ Investment increases somewhat over the first three quarters, but then declines until it reaches a trough located about 30% below trend in quarter 11 (2009Q4). By contrast, credit to nonfinancial businesses substantially increased

23. Recall that the passive banking sector in the BGG model does not have any effect on the real dynamics of the economy and in any case has been added for purely pedagogic purposes.

24. The quadratic trends were estimated over the period 1960Q1–2013Q4.

relative to trend until peaking at 7.5% above trend in the fifth quarter (2008Q2), and fall below their 2007Q1 value only in quarter 11, followed by a persistent decline. The bank capital ratio gradually declines until it is almost 1 percentage point below its steady state in quarter 7 (2008Q4), and then quickly recovers and moves substantially above its precrisis value. Finally, our measure of the cost of external finance increases continuously until peaking at almost 7.5% in the seventh quarter (2008Q4).²⁵

The full model reproduces about two-thirds of the trough of investment and about three quarters of the peak of the cost of external finance, respectively. Most likely due to the absence of any frictions constraining the ability of households to borrow, the model fails to reproduce the decline in consumption. As a result, it reproduces about half of the trough of GDP. The model also closely matches the path of the bank capital ratio. Finally, it does not reproduce the observed increase in the debt of nonfinancial firms during the course of 2008. This increase has been attributed by Ivashina and Scharfstein (2010) as well as Adrian, Colla, and Shin (2012) to a precautionary take-down of credit lines by firms concerned about the solvency and liquidity of the banking sector. Such motives for credit demand are absent from the model.

By contrast, the BGG model reproduces less than one-half of the peak of the external finance premium and only about a fifth of the trough of investment and generates almost no decline in consumption. The GK model is more successful than the BGG model at reproducing the observed decline in investment, though less so than the full model. Furthermore, the GK model generates an *increase* in consumption and a substantially too low path for the bank capital ratio.²⁶

6. CONCLUSION

I develop a general equilibrium model combining informational frictions between banks and entrepreneurs as well as banks and depositors. I do so by adding a moral hazard problem between banks and depositors along the lines of Gertler and Karadi (2011) to the Bernanke, Gertler, and Gilchrist (1999) financial accelerator model. As a result, both entrepreneurial and bank leverage matter for the cost of external funds of firms. I compare my model to a BGG-type financial accelerator model and a Gertler–Karadi-type model. I find that adding the friction between banks and depositors amplifies the response of the cost of external finance and the overall economy to monetary policy and productivity shocks as compared to a BGG-type financial accelerator model. The additional amplification provided by this “bank capital channel” allows my model to improve upon the BGG model’s ability to match the volatility of the cost of external finance in the data, as well as investment and

25. The alternative measures of the cost of external finance mentioned in footnote 11 closely track this measure.

26. The latter remains true if we employ the “Fair Value” method of calculating bank assets and equity suggested by Gertler and Karadi (2013).

other variables. Moreover, in the full model, bank leverage declines in response to contractionary monetary policy and productivity shocks, which allows the full model to match the procyclicality of bank leverage in U.S. data. By contrast, bank leverage in the Gertler–Karadi-type model is strongly countercyclical.

Furthermore, an adverse shock to entrepreneurial net worth causes an output contraction more than twice as large as in a BGG-type model. In line with the existing empirical evidence, an adverse shock to bank net worth causes a persistent decline in GDP. For a reasonably calibrated combination of both balance sheet shocks, the model economy displays a contraction of investment and an increase in the cost of external finance of magnitudes similar to the experience of the U.S. economy during the Great Recession associated with the financial crisis of 2007–09.

APPENDIX A: FOCS OF HOUSEHOLDS, RETAILERS, CAPITAL GOODS' PRODUCERS, BANKERS, AND ENTREPRENEURS

The economy features a representative household with preferences described by the intertemporal utility function

$$E_t \left\{ \sum_{i=0}^{\infty} \beta^i \left[\ln(C_{t+i} - hC_{t+i-1}) - \frac{\chi}{1+\varphi} (l_{t+i}^s)^{1+\varphi} \right] \right\},$$

where C_t and l_t^s denote a CES basket of consumption good varieties and labor effort, respectively, and h denotes the degree of external habit formation. In period t , the household saves by depositing funds with banks, denoted as B_t and by buying government bonds, denoted as B_t^g . Both of these assets have a maturity of one-quarter, yield a nominal return and, in the equilibrium considered here, are perfectly risk-free in nominal terms. Households also earn wage income from supplying labor to retailers and derive profit income from their ownership of retail firms and capital goods producers. Hence, their budget constraint is given by

$$P_t C_t = w_t P_t l_t + P_t \text{prof}_t + R_{t-1} B_{t-1} - B_t + R_t^{gov} B_{t-1} - B_t^b, \quad (\text{A1})$$

where C_t , w_t , and prof_t denote consumption, the real wage, and real profits, respectively.

The first-order conditions with respect to consumption, riskless assets (i.e., deposits and bonds), and labor l_t are given by

$$q_t = \frac{1}{C_t - hC_{t-1}} - \beta h E_t \left\{ \frac{1}{C_{t+1} - hC_t} \right\}, \quad (\text{A2})$$

$$q_t = \beta E_t \left[q_{t+1} \frac{R_t}{\Pi_{t+1}} \right], \quad (\text{A3})$$

$$Q_t = \beta E_t \left[Q_{t+1} \frac{b_t R_t^{gov}}{\Pi_{t+1}} \right], \quad (A4)$$

$$Q_t w_t = \chi(l_t)^\phi, \quad (A5)$$

where $\Pi_t = \frac{P_t}{P_{t-1}}$. Hence $R_t = R_t^{gov}$.

In the model without a financial sector, I assume that households buy the capital stock K_t from capital goods producers in order to rent it out to retailers in period $t + 1$. Hence, the budget constraint becomes

$$P_t C_t + Q_t (K_t - K_{t-1}) \quad (A6)$$

$$= P_t l_t w_t + P_t profit_t + R_{t-1} B_{t-1}^T - B_t^T. \quad (A7)$$

This modification leaves the first-order conditions derived above unchanged, but adds a first-order condition with respect to K_t :

$$Q_t = E_t \left\{ \beta \frac{Q_{t+1}}{Q_t} [r_{t+1}^K + Q_{t+1}(1 - \delta)] \right\}. \quad (A8)$$

A.1 Capital Goods' Producers

The first-order condition with respect to I_t is given by

$$\begin{aligned} Q_t \left(1 - \frac{\eta_i}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \right) &= 1 + Q_t \eta_i \left(\frac{I_t}{I_{t-1}} - 1 \right) \frac{I_t}{I_{t-1}} \\ &- E_t \left\{ \beta \frac{Q_{t+1}}{Q_t} Q_{t+1} \eta_i \left(\frac{I_{t+1}}{I_t} - 1 \right) \left(\frac{I_{t+1}}{I_t} \right)^2 \right\}. \end{aligned} \quad (A9)$$

The law of motion of capital is given by

$$K_t = (1 - \delta)K_{t-1} + I_t \left(1 - \frac{\eta_i}{2} \left(\frac{I_t}{I_{t-1}} - 1 \right)^2 \right). \quad (A10)$$

A.2 Retailers

Cost minimization, the assumption of economy wide-factor markets, and $\int K_t^S(i) = K_{t-1}$ imply that

$$w_t(1 + \psi_l(R_t - 1)) = (1 - \alpha)mc_t \frac{Y_t}{l_t}, \quad (A11)$$

$$r_t^k(1 + \psi_K(R_t - 1)) = \alpha mc_t \frac{Y_t}{K_{t-1}}, \quad (\text{A12})$$

$$L_t^r = \psi_L w_t l_t + \psi_K r_t^k K_{t-1}, \quad (\text{A13})$$

where mc_t denotes the real marginal cost of production.

Retail firms are subject to nominal rigidities in the form of Calvo (1983) contracts: Only a fraction $1 - \xi$ is allowed to optimize its price in a given period. Those firms that are not allowed to optimize their prices index them to past inflation at a rate of γ_P and to the steady-state inflation rate Π at rate $1 - \gamma_P$. The firm's problem is then to choose $p_t(i)$ in order to maximize

$$E_t \left\{ \sum_{i=0}^{\infty} (\xi^P \beta)^i \frac{Q_{t+i}}{Q_t} \left[\left(\frac{p_t(i)}{P_{t+i}} \prod_{k=1}^i \Pi^{1-\gamma_P} \Pi_{t+k}^{\gamma_P} \right)^{1-\varepsilon} - mc_{t+i} \left(\frac{p_t(i)}{P_{t+i}} \prod_{k=1}^i \Pi^{1-\gamma_P} \Pi_{t+k}^{\gamma_P} \right)^{-\varepsilon} \right] Y_{t+i} \right\}.$$

The first-order condition is given by

$$\tilde{p}_t = \frac{\varepsilon}{\varepsilon - 1} \frac{E_t \left\{ \sum_{i=0}^{\infty} (\xi^P \beta)^i \frac{Q_{t+i}}{Q_t} \left(\frac{\prod_{k=1}^i \Pi^{1-\gamma_P} \Pi_{t+k}^{\gamma_P}}{\prod_{k=1}^i \Pi_{t+k}} \right)^{-\varepsilon} mc_{t+i} Y_{t+i} \right\}}{E_t \left\{ \sum_{i=0}^{\infty} (\xi^P \beta)^i \frac{Q_{t+i}}{Q_t} \left(\frac{\prod_{k=1}^i \Pi^{1-\gamma_P} \Pi_{t+k}^{\gamma_P}}{\prod_{k=1}^i \Pi_{t+k}} \right)^{1-\varepsilon} Y_{t+i} \right\}}, \quad (\text{A14})$$

with $\tilde{p}_t = \frac{p_t^*}{P_t}$, where p_t^* denotes the price chosen by those firms that are allowed to optimize in period t . The law of motion of the price index is given by

$$P_t = \left[(1 - \xi^P) (p_t^*)^{1-\varepsilon} + \xi^P (P_{t-1} \Pi^{1-\gamma_P} (\Pi_{t-1})^{\gamma_P})^{1-\varepsilon} \right]^{\frac{1}{1-\varepsilon}}. \quad (\text{A15})$$

A.3 Bankers

Combining (3) with the definition of $V_t^b(q)$ allows the latter to be expressed as

$$V_t^b(q) = v_t L_t^e(q) + \eta_t N_t^b(q), \quad (\text{A16})$$

with

$$v_t = E_t \left\{ (1 - \theta) \frac{(R_{t+1}^b - R_t)}{R_t} + \frac{\theta x_{t,t+1} v_{t+1} \Pi_{t+1}}{R_t} \right\}, \quad (\text{A17})$$

$$\eta_t = E_t \left\{ (1 - \theta) + \frac{\theta z_{t,t+1} \eta_{t+1} \Pi_{t+1}}{R_t} \right\}, \quad (\text{A18})$$

$$x_{t,t+1} = \frac{L_{t+1}^e(q)}{L_t^e(q)}, \quad z_{t,t+1} = \frac{N_{t+1}^b(q)}{N_t^b(q)}. \quad (\text{A19})$$

Using $V_t^b(q) = \lambda L_t^e(q)$ yields

$$L_t^e(q) = \phi_t^b N_t^b(q), \quad (\text{A20})$$

$$\phi_t^b(q) = \frac{\eta_t}{\lambda - v_t}, \quad (\text{A21})$$

where $\phi_t^b(q)$ denotes bank q 's leverage ratio. Note that a necessary condition for the incentive constraint to bind is $0 < v_t < \lambda$.²⁷ Substituting (3) and (A20) into (A19) allows $z_{t,t+1}$ and $x_{t,t+1}$ to be written as

$$z_{t,t+1} = \frac{(R_{t+1}^b - R_t) \phi_t^b(q) + R_t}{\Pi_{t+1}} \exp(e_{t+1}^z), \quad (\text{A22})$$

$$x_{t,t+1} = \frac{\phi_{t+1}^b(q)}{\phi_t^b(q)} z_{t,t+1}. \quad (\text{A23})$$

Equations (A21), (A17), (A18), (A22), and (A23) imply that η_t , v_t , $\phi_t^b(q)$, $z_{t,t+1}$, and $x_{t,t+1}$ depend solely on economy-wide variables and $\phi_{t+1}^b(q)$, implying that they all depend on economy-wide variables alone. This allows for easy aggregation across bankers, implying that

$$L_t^e = \phi_t^b N_t^b, \quad (\text{A24})$$

where N_t^b denotes the total net worth of banks.

27. For an interpretation of this condition, see Gertler and Karadi (2011).

A.4 Entrepreneurs

Using $\bar{\omega}_{t+1}^j R_{t+1}^K P_t Q_t K_t^j = R_t^L P_t L_t^j$ and $P_t L_t^j = P_t Q_t K_t^j - P_t N_t^j$, rewrite the participation constraint of the bank (11) as

$$\left(P_t Q_t K_t^j - P_t N_t^j \right) E_t R_{t+1}^b = E_t \left\{ R_{t+1}^K P_t Q_t K_t^j \left[\bar{\omega}_{t+1}^j \int_{\bar{\omega}_{t+1}^j}^{\infty} f(\omega^j) d\omega^j + (1 - \mu) \int_0^{\bar{\omega}_{t+1}^j} \omega^j f(\omega^j) d\omega^j \right] \right\}$$

or

$$\left(\phi_t^e(j) - 1 \right) E_t R_{t+1}^b = \phi_t^{e,j} E_t \left\{ R_{t+1}^K \left[\Gamma \left(\bar{\omega}_{t+1}^j \right) - \mu G \left(\bar{\omega}_{t+1}^j \right) \right] \right\}, \quad (\text{A25})$$

where $\phi_t^e(j) = \frac{Q_t K_t^j}{N_t^j}$, $\Gamma(\bar{\omega}_{t+1}^j) = \bar{\omega}_{t+1}^j \int_{\bar{\omega}_{t+1}^j}^{\infty} f(\omega^j) d\omega^j + \int_0^{\bar{\omega}_{t+1}^j} \omega^j f(\omega^j) d\omega^j$, and $G(\bar{\omega}_{t+1}^j) = \int_0^{\bar{\omega}_{t+1}^j} \omega^j f(\omega^j) d\omega^j$. Using $\bar{\omega}_{t+1}^j R_{t+1}^K Q_t K_t^j = R_t^L L_t^j$ and $E(\omega_{t+1}) = 1 = \int_0^{\bar{\omega}_{t+1}^j} \omega^j f(\omega^j) d\omega^j + \int_{\bar{\omega}_{t+1}^j}^{\infty} \omega^j f(\omega^j) d\omega^j$, rewrite the entrepreneur's objective as

$$\begin{aligned} & P_t Q_t K_t^j E_t \left\{ \int_{\bar{\omega}_{t+1}^j}^{\infty} R_{t+1}^K f(\omega^j) \left(\omega^j - \bar{\omega}_{t+1}^j \right) d\omega^j \right\} \\ &= P_t Q_t K_t^j E_t \left\{ R_{t+1}^K \left[1 - \int_0^{\bar{\omega}_{t+1}^j} \omega^j f(\omega^j) d\omega^j - \bar{\omega}_{t+1}^j \int_{\bar{\omega}_{t+1}^j}^{\infty} f(\omega^j) d\omega^j \right] \right\} \\ &= P_t Q_t K_t^j E_t \left\{ R_{t+1}^K \left[1 - \Gamma \left(\bar{\omega}_{t+1}^j \right) \right] \right\} \\ &= \phi_t^e(j) E_t \left\{ R_{t+1}^K \left[1 - \Gamma \left(\bar{\omega}_{t+1}^j \right) \right] \right\} N_t^j. \end{aligned}$$

Recall that entrepreneurs differ only in their net worth N_t^j . Since $\bar{\omega}_{t+1}^j = \frac{R_t^L (1 - \frac{1}{\phi_t^e(j)})}{R_{t+1}^K}$, the values of $\phi_t^e(j)$ and R_t^L maximizing $\phi_t^e(j) E_t \{ R_{t+1}^K [1 - \Gamma(\bar{\omega}_{t+1}^j)] \} N_t^j$ subject to (A25) will be the same across entrepreneurs. The same is true of the cutoff value $\bar{\omega}_{t+1}^j$. Hence, the entrepreneur's problem is to maximize

$$\begin{aligned} & \phi_t^e E_t \left\{ R_{t+1}^K [1 - \Gamma(\bar{\omega}_{t+1})] \right\} + \xi_t E_t \left\{ \phi_t^e R_{t+1}^K [\Gamma(\bar{\omega}_{t+1}) - \mu G(\bar{\omega}_{t+1})] \right. \\ & \quad \left. - R_{t+1}^b (\phi_t^e - 1) \right\}. \end{aligned}$$

The first-order conditions with respect to ϕ_t^e , R_t^L , and ξ_t are given by

$$E_t \left\{ R_{t+1}^K [1 - \Gamma(\bar{\omega}_{t+1})] \right\} + \xi_t E_t \left\{ R_{t+1}^K [\Gamma(\bar{\omega}_{t+1}) - \mu G(\bar{\omega}_{t+1})] - R_{t+1}^b \right\} = 0, \quad (\text{A26})$$

$$E_t \{-\Gamma'(\bar{\omega}_{t+1}) + \xi_t [\Gamma'(\bar{\omega}_{t+1}) - \mu G'(\bar{\omega}_{t+1})]\} = 0, \quad (\text{A27})$$

$$E_t \{\phi_t^e R_{t+1}^K [\Gamma(\bar{\omega}_{t+1}) - \mu G(\bar{\omega}_{t+1})] - R_{t+1}^b (\phi_t^e - 1)\} = 0, \quad (\text{A28})$$

where ξ_t denotes the Lagrange multiplier on the banks' participation constraint. Given that we can rewrite the entrepreneur's objective as we have, both in the full model and in the BGG model, total real entrepreneurial equity at the beginning of period t (i.e., before some entrepreneurs die) V_t is given by

$$V_t = Q_{t-1} K_{t-1} \frac{R_t^K}{\Pi_t} [1 - \Gamma(\bar{\omega}_t)] \exp(e_t^N). \quad (\text{A29})$$

In the BGG model, the constraint on the return on the portfolio of loans to entrepreneurs holds not just in expectation, but in every $t + 1$ state. Furthermore, due to the absence of a moral hazard problem in the banking sector, the return on bank loans made in period t equals the deposit rate. Hence, the bank's participation constraint is given by

$$\phi_t^e R_{t+1}^K [\Gamma(\bar{\omega}_{t+1}) - \mu G(\bar{\omega}_{t+1})] - R_t (\phi_t^e - 1) = 0. \quad (\text{A30})$$

The optimization problem then becomes to maximize

$$\phi_t^e E_t \{R_{t+1}^K [1 - \Gamma(\bar{\omega}_{t+1})]\} + \xi_t [\phi_t^e R_{t+1}^K [\Gamma(\bar{\omega}_{t+1}) - \mu G(\bar{\omega}_{t+1})] - R_t (\phi_t^e - 1)]$$

with respect to ϕ_t^e , $\bar{\omega}_{t+1}$, and ξ_t . The first-order conditions are

$$E_t \{R_{t+1}^K [1 - \Gamma(\bar{\omega}_{t+1})] + \xi_t \{R_{t+1}^K [\Gamma(\bar{\omega}_{t+1}) - \mu G(\bar{\omega}_{t+1})] - R_t\}\} = 0, \quad (\text{A31})$$

$$\frac{\Gamma'(\bar{\omega}_{t+1})}{[\Gamma'(\bar{\omega}_{t+1}) - \mu G'(\bar{\omega}_{t+1})]} = \xi_t, \quad (\text{A32})$$

$$R_t^K [\Gamma(\bar{\omega}_t) - \mu G(\bar{\omega}_t)] = R_{t-1} \frac{(\phi_{t-1}^e - 1)}{\phi_{t-1}^e}. \quad (\text{A33})$$

The loan rate in this setup is only determined once the loan is paid back, in order to ensure that (A30) holds. It is determined by

$$\frac{R_t^L (Q_{t-1} K_{t-1} - N_{t-1})}{R_t^K Q_{t-1} K_{t-1}} = \bar{\omega}_t. \quad (\text{A34})$$

Finally, by combining (A29) and (A33), as well as (14), entrepreneurial net worth can be shown to evolve as

$$N_t = \exp(e_t^N) \gamma \left[Q_{t-1} K_{t-1} \frac{R_t^K}{\Pi_t} (1 - \mu G(\bar{\omega}_t)) - \frac{R_{t-1}}{\Pi_t} (Q_{t-1} K_{t-1} - N_{t-1}) \right] + W^e. \quad (\text{A35})$$

APPENDIX B: DATA SOURCES

All components of GDP were divided by the labor force. Except the interest rates or interest rate spreads, all data series reported in nominal terms by the respective data provider were deflated using the GDP deflator and divided by the labor force.

- Population level: Bureau of Labor Statistics (BLS) “Civilian noninstitutional population” series, ID LNS10000000Q. Following Smets and Wouters (2007), for years before 1976, I used LFU800000000 : Population level—16 Years and Older.
- GDP deflator: BEA, NIPA Table 1.1.9. Implicit Price Deflators for Gross Domestic Product.
- Real Gross Domestic Product, quantity index, seasonally adjusted: Bureau of Economic Analysis (BEA), NIPA Table 1.1.3.
- Real Personal Consumption Expenditures, quantity index, seasonally adjusted, BEA, NIPA Table 1.1.3.
- Real Gross Private Domestic Investment—Fixed Nonresidential, Quantity Index, seasonally adjusted, BEA, NIPA Table 1.1.3.
- Hours worked: Hours worked in the nonfarm business sector, Bureau of labor statistics, seasonally adjusted, PRS85006033
- Net worth of entrepreneurs: FFA of the Federal Reserve Board, sum of “Non-financial corporate business; net worth,” series ID FL102090005.Q, and “Non-financial noncorporate business; proprietors’ equity in noncorporate business,” series ID FL112090205.Q. The series was seasonally adjusted.
- Leverage of the nonfinancial sector: FFA. The numerator is the sum of “Nonfinancial corporate business; total assets,” series ID FL102000005.Q and “Nonfinancial noncorporate business; total assets,” series ID FL112000005.Q. The denominator is “Net worth of entrepreneurs” as described above. The resulting series was seasonally adjusted.
- Loans to nonfinancial businesses: FFA. Sum of “Nonfinancial corporate business; credit market instruments,” series ID FL104104005.Q, and “Nonfinancial noncorporate business; credit market instruments,” series ID FL114104005.Q. “Credit market instruments” consists of six debt instruments: commercial paper, municipal securities, corporate bonds, depository institution loans not elsewhere

classified, other loans, and advances and mortgages. The resulting series was seasonally adjusted.

- Net worth of banks: FDIC. TCE, calculated using the FDIC's "Quarterly Banking Profile" (QBP), table "Assets and Liabilities of FDIC-Insured Commercial Banks and Savings Institutions." TCE is calculated as total equity capital—Perpetual preferred stock—Intangible assets. The resulting series was seasonally adjusted.
- Capital ratio: The numerator is TCE. The denominator is "Risk-Weighted Assets" of FDIC-Insured Commercial Banks and Savings Institutions, available from the FDIC on request. The resulting series was seasonally adjusted.
- Bank profits: FDIC, calculated using the FDIC's "Quarterly Banking Profile" (QBP), table Quarterly Income and Expense of FDIC-Insured Commercial Banks and Savings Institutions. Profits are calculated as the sum of "Net income (loss) attributable to bank" and "Applicable income taxes."
- Cost of external finance: Federal Reserve Bank of St. Louis, Moody's Seasoned Baa Corporate Bond Yield-Effective Federal Funds Rate. Alternative measures: Moody's Seasoned Baa Corporate Bond Yield—3-month treasury bill rate, Moody's Seasoned Aaa Corporate Bond Yield-Effective Federal Funds Rate, Moody's Seasoned Aaa Corporate Bond Yield—3-month treasury bill rate.

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